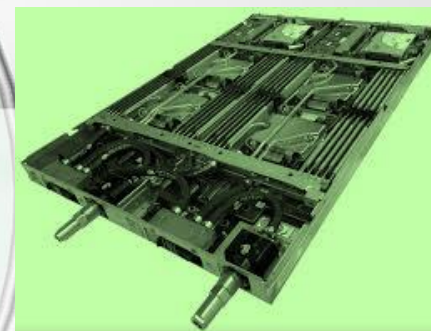


液冷時代來臨：資料中心與高功率電子的 熱管理革新

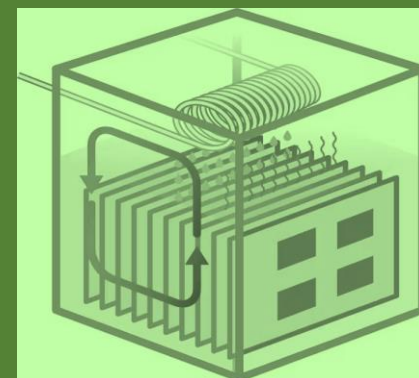
劉君愷

台灣熱管理協會



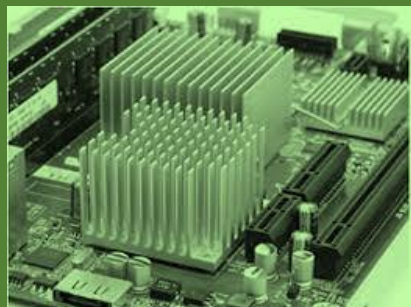
內容

- 台灣熱管理產業發展
- AI伺服器發展趨勢與散熱問題
- 液冷散熱技術發展
- 台灣產業機會
- 結論



台灣熱管理協會

- 於96年10月成立



TTMA 台灣熱管理協會
Taiwan Thermal Management Association

首頁 網站地圖 English

關於協會 會員專區 活動訊息 出版刊物 相關網站 聯絡我們

台灣熱管理協會
Taiwan Thermal Management Association

協、會、簡、介
Introduction

本會為依法設立，非以營利為目的之社會團體。係以結合電子散熱產業相關之產、官、學、研界共同促進熱管理產業及技術發展為宗旨。

快、速、選、單
Quick Menu

加入會員 線上報名 出版品

AWPA 先進微系統與機裝技術聯盟

台灣顯示器材料與元件產業協會
TMDA Taiwan Display Materials & Devices Association

最新活動 MORE ▶

- 摘要表截止時間延長至2/21(五)前~2025年會暨技術成果發表... 2025/02/21
| 舉辦時間：2025/4/
| 舉辦地點：張榮發基金會國際會議中心
- 氣冷散熱技術研討會 2024/10/31
| 舉辦時間：2024/11/13 (三)
| 舉辦地點：嘉思台大會議中心 柏拉圖廳

最新公告 MORE ▶

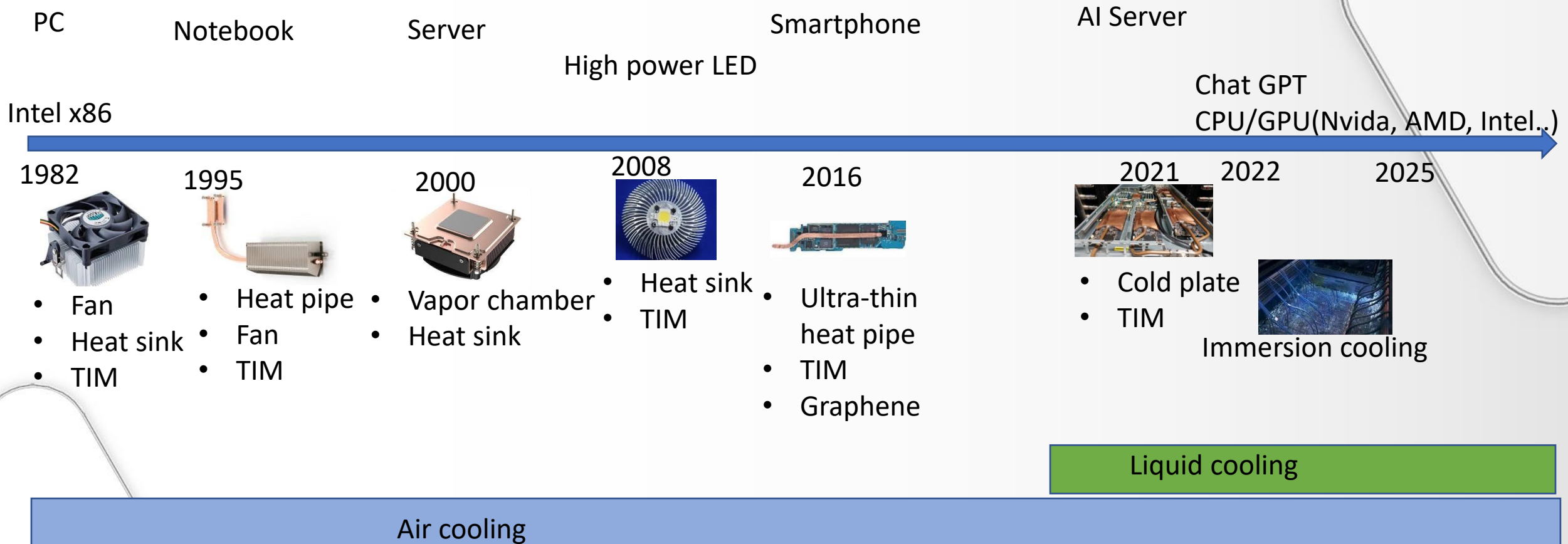
產業要聞 MORE ▶

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<http://www.thermal.org.tw/>

電子散熱發展趨勢

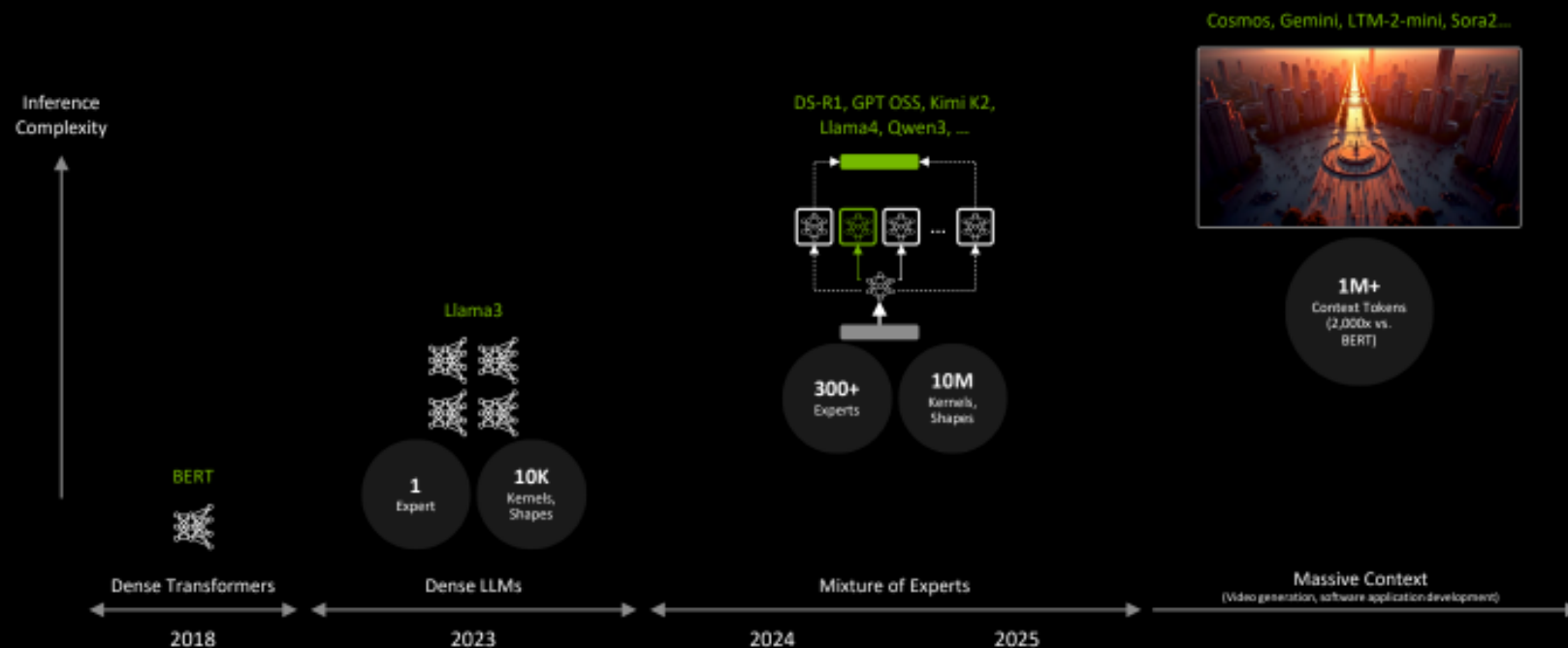
- 散熱產業促成了各式重要電子產品的成長



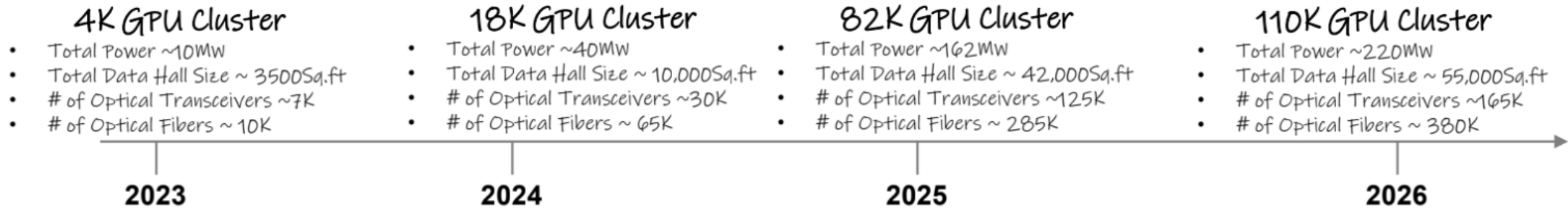
AI 推論複雜度爆發

Inference Complexity is Exploding

More parameters, experts, reasoning, kernels & shapes, and context



AI 運算叢集增加



Giga-Scale AI 正轉移到Data Centers

Driving extreme co-design from chip to grid with open collaboration

Compute

Networking

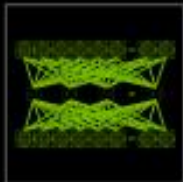
Mechanical

Power

Cooling



NVIDIA Giga-Scale Reference Designs



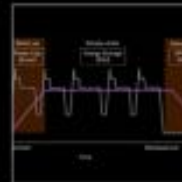
Scale-Up



CPX



Spectrum-X
Ethernet



Power
Smoothing



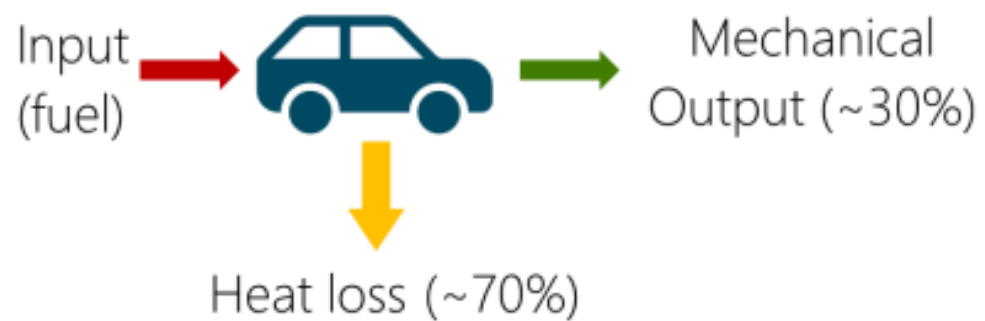
45°C
Liquid Cooling



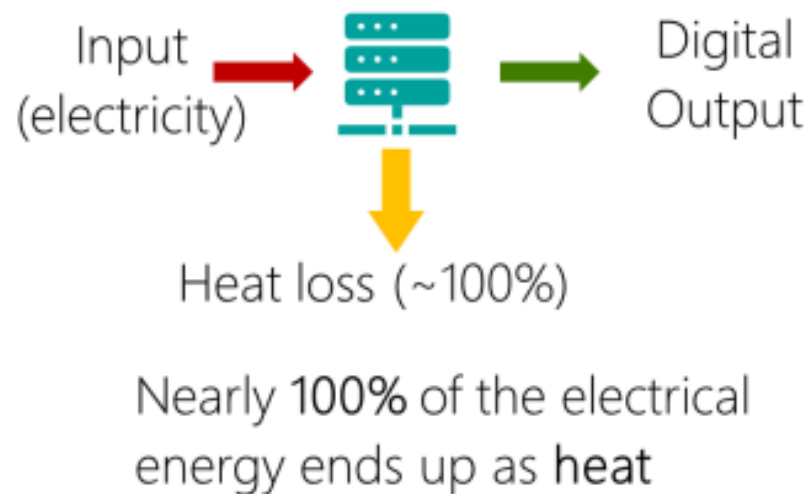
Open
Collaboration

資料中心能量轉換

Internal Combustion Engine



Server in a Data Center

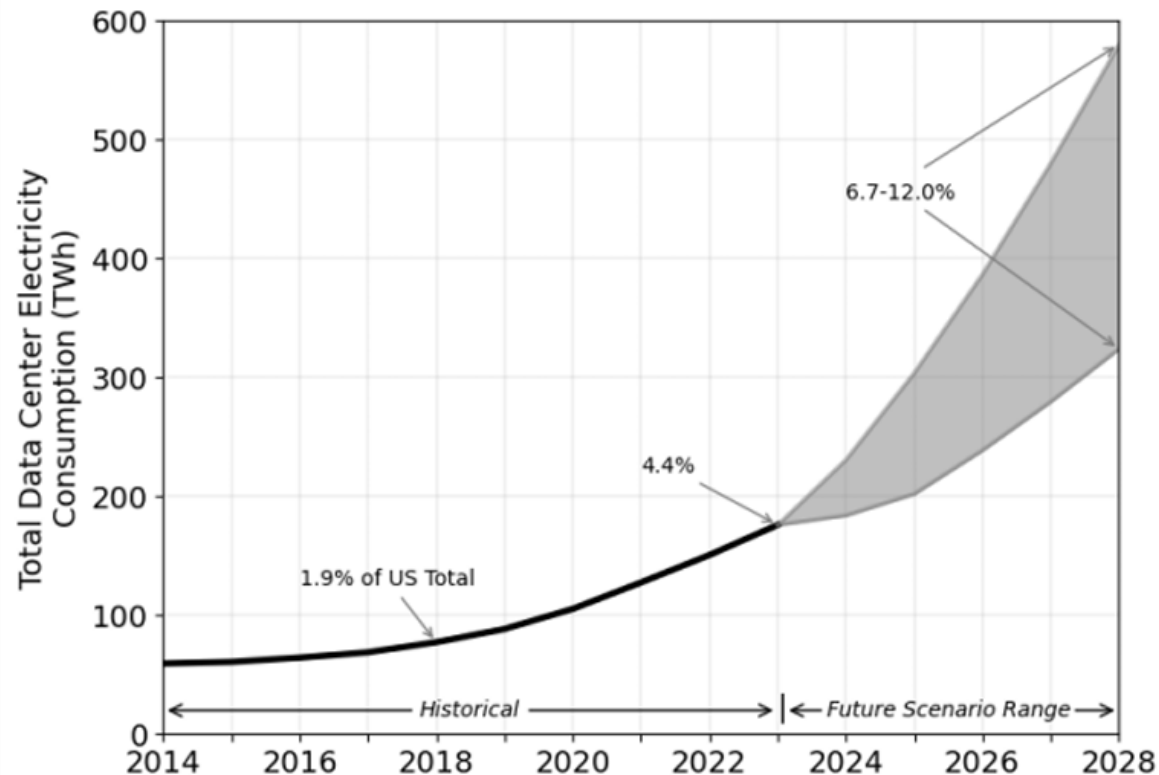


資料中心電力消耗成長

2023年，美國資料中心消耗了176太瓦時（TWh）電力，佔總電力消耗的4.4%。

- 176太瓦時的電力可供4000萬美國家庭使用一年。
- 預計2028年，電力消耗將達到325至580太瓦時。
- 受人工智慧、雲端運算和工業擴張的驅動。在全球範圍內，資料中心產業

預計到2030年將排放25億噸二氧化碳。



<https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report.pdf>

資料中心過熱問題

Serious consequences affecting hardware and operation

Hardware
damage

Performance
slowdown

System
outages and
data loss

Reduced
Equipment
lifespan

Fire risk

Downtime
and financial
loss

September 2024:

A fire at a data center in Singapore caused a Chinese CSP outage

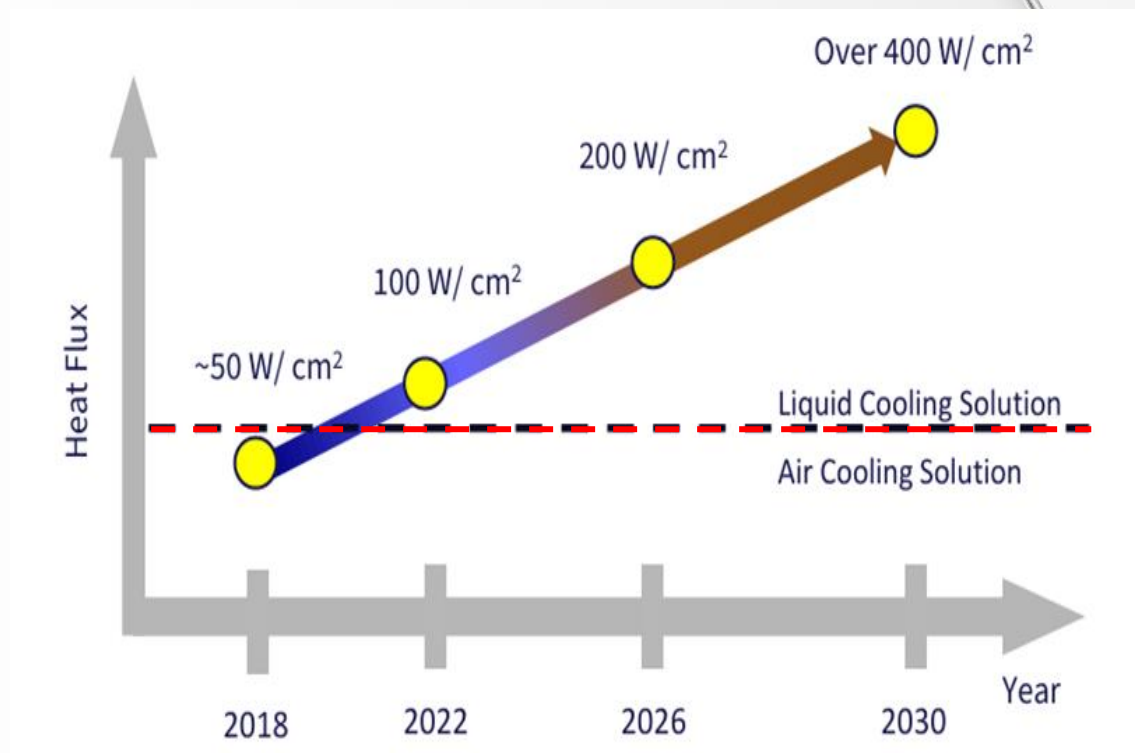
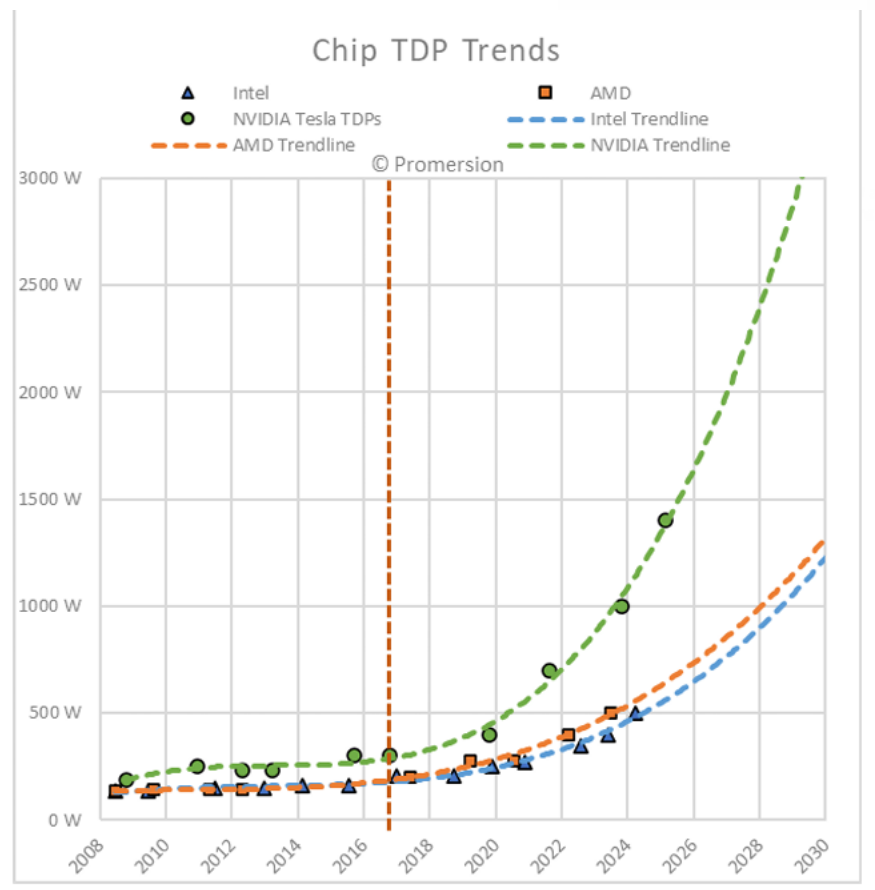
August 2024:

A fire broke out at the 350 East Cermak data center in Chicago

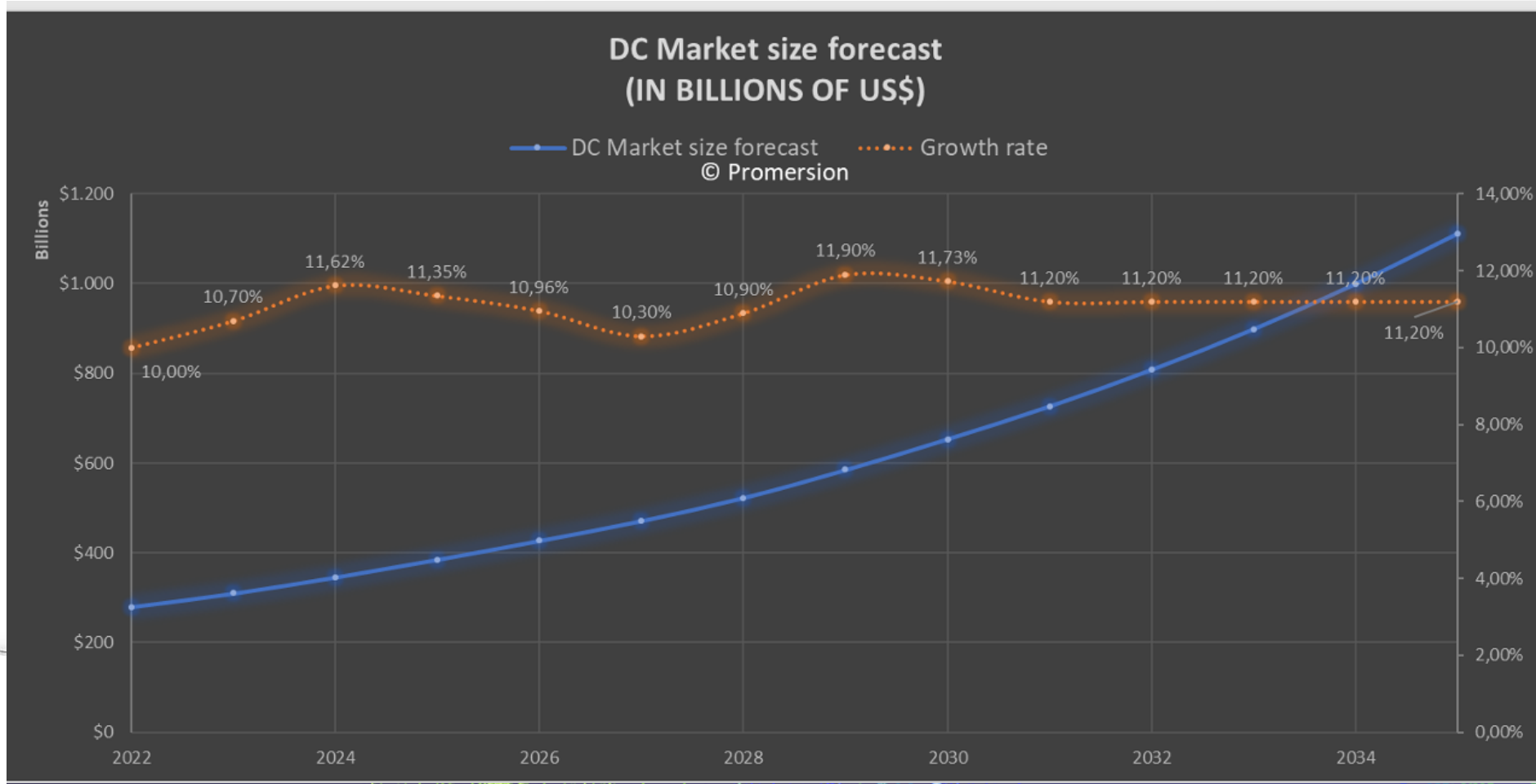
2025:

A fire occurred in one of the racks at the data center in São Paulo, impacting a data hall

資料中心發熱趨勢

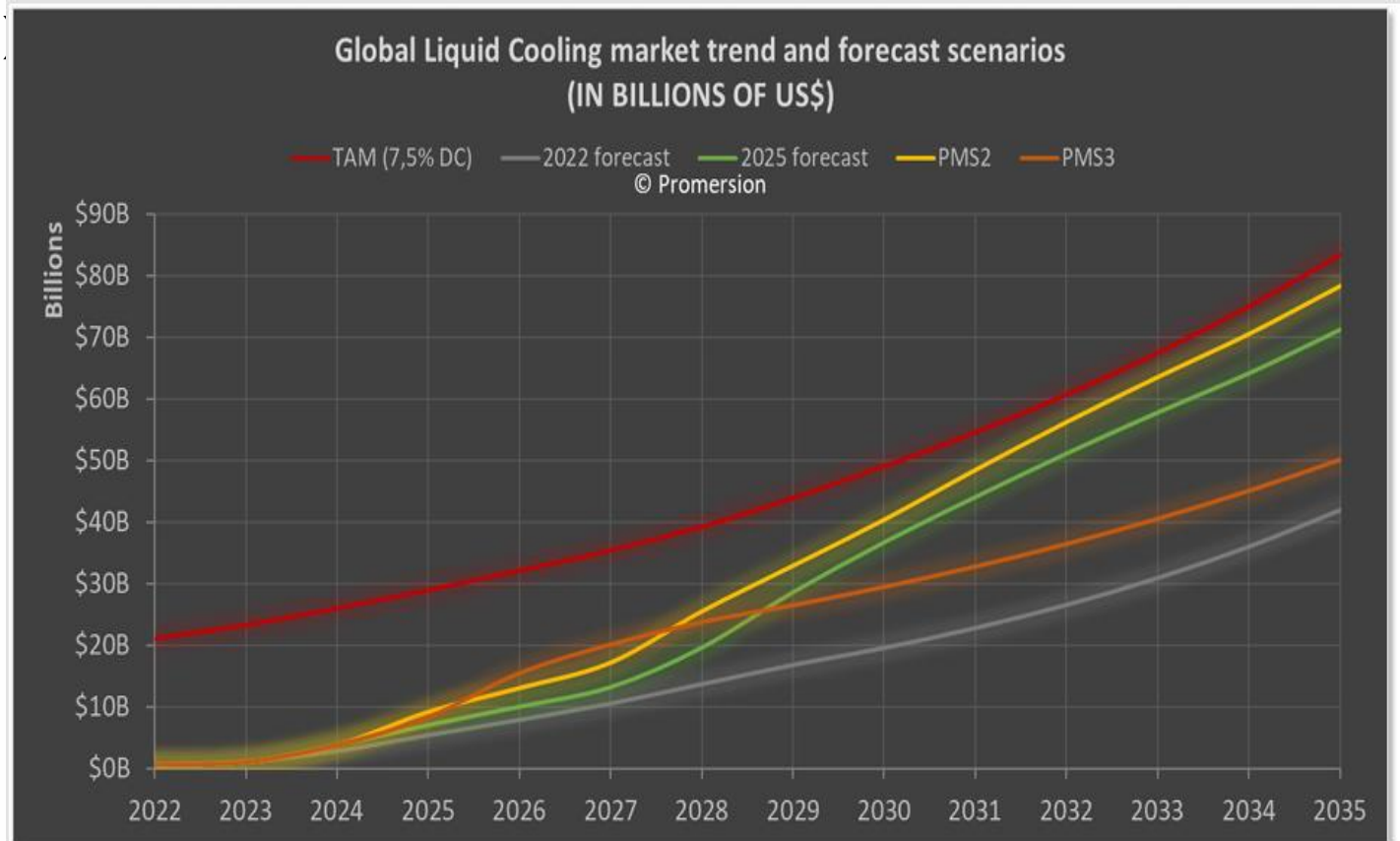


資料中心市場



全球液冷市場需求

- 「總可用市場」(Total Addressable Market) 假設資料中心 (Data Center) 整體以 7.5% 的年複合成長率。到 2035 年會超過 80-85B 美元。
- 2022 年的舊預測成長速度明顯較慢，大約到 2035 年落在 40B 美元附近。
- 2025 更新預測，顯示市場會比 2022 年的預測 更快成長。2035 年接近 70B 美元。
- 黃色線：PMS2 中高成長情境。約在 75B 美元左右。
- PMS3 較保守但還算樂觀的情境。2035 年落在約 50B 美元附近。



散熱需求造成技術轉折點

Power delivery to CPU/GPU:

- Main losses:
- Voltage Regulation Module (VRM) losses: 85-95% efficiency
- Power Delivery Network (PDN) and other losses: 97%-99% efficiency
- Per 500W CPU: 66W
- Per 1200W GPU: 157W

Network cards

- High Bandwidth Ethernet NICs: up-to 60W/port
- **SmartNICs: up-to 150W/port**
- Infiniband: up-to 100W
- **FPGA SmartNICs: up-to 250W**

Memory (per CPU)

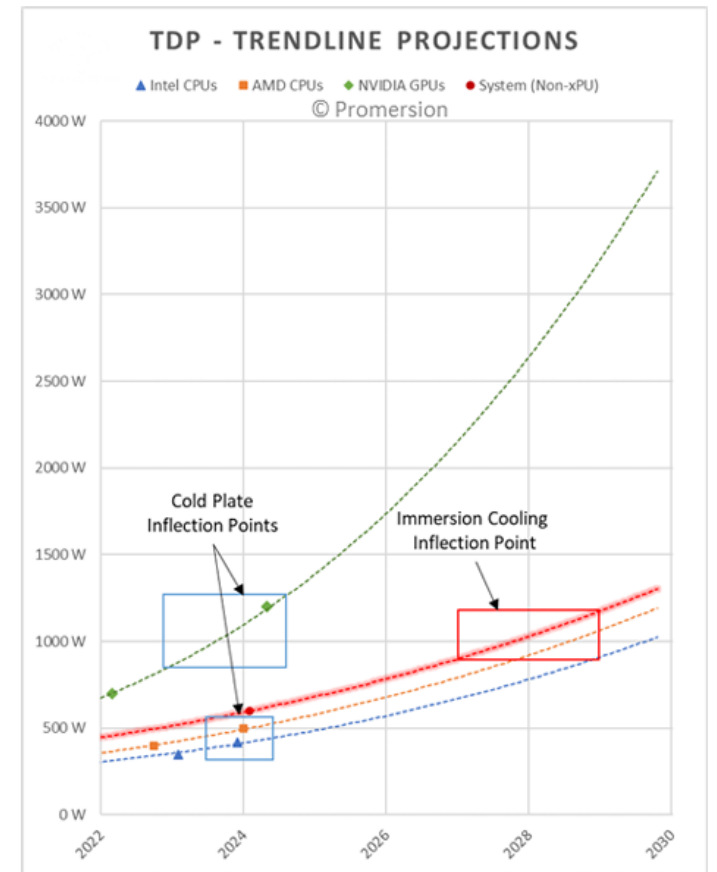
- **DDR: 12-16x 20 W = up-to 240-320W**
- HBM: up-to 100W/module

Storage

- **Storage accelerator: up-to 70W**
- **NVMe SSD: up-to 30W /drive**

Power Supply

- 80 PLUS Platinum PSU's
- 94-98% efficiency
- Losses of +/- 4% of system power
- Up-to 40W per kW total system power



液冷市場預估

冷板

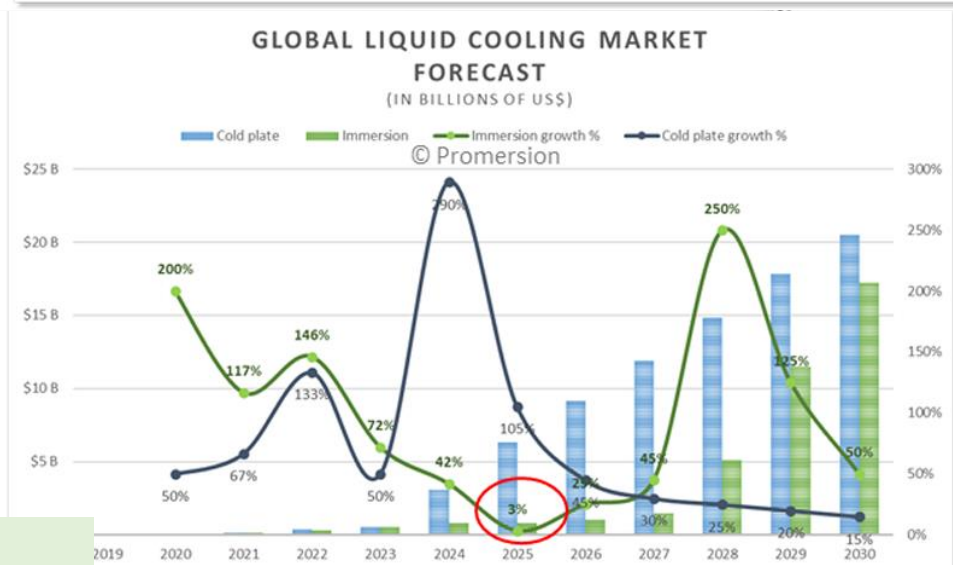
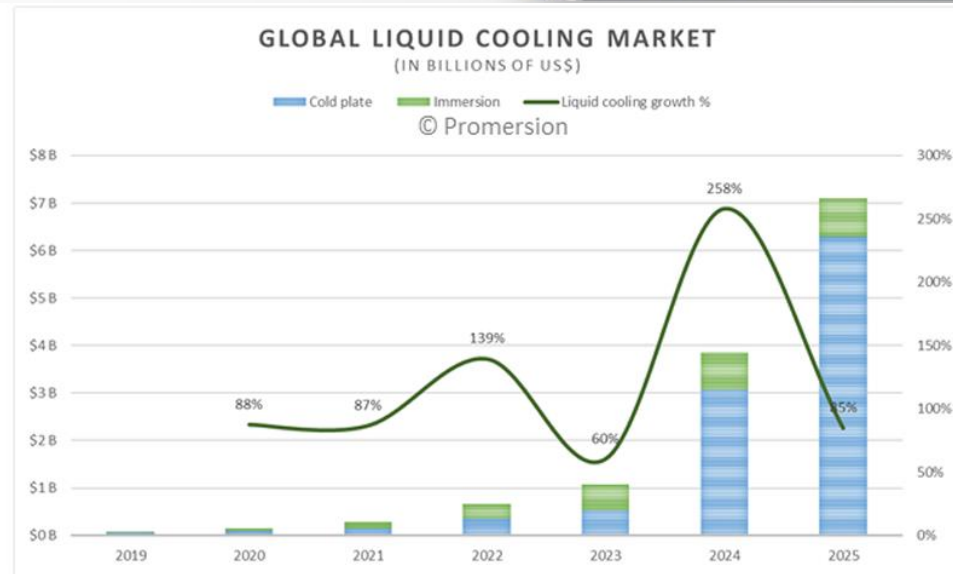
- 由於散熱需求，冷板將在 **2024/2025** 年迎來爆炸性成長
- 2025 年進展順利，供應鏈快速擴張
- 長期市場成長伴隨著成本降低，導致單價收入下降，銷量成長抑制了營收成長
- 冷板正逐漸成為標準配置，並越來越多地整合到熱封裝中，從而抑制了獨立冷板技術的發展

浸沒式散熱

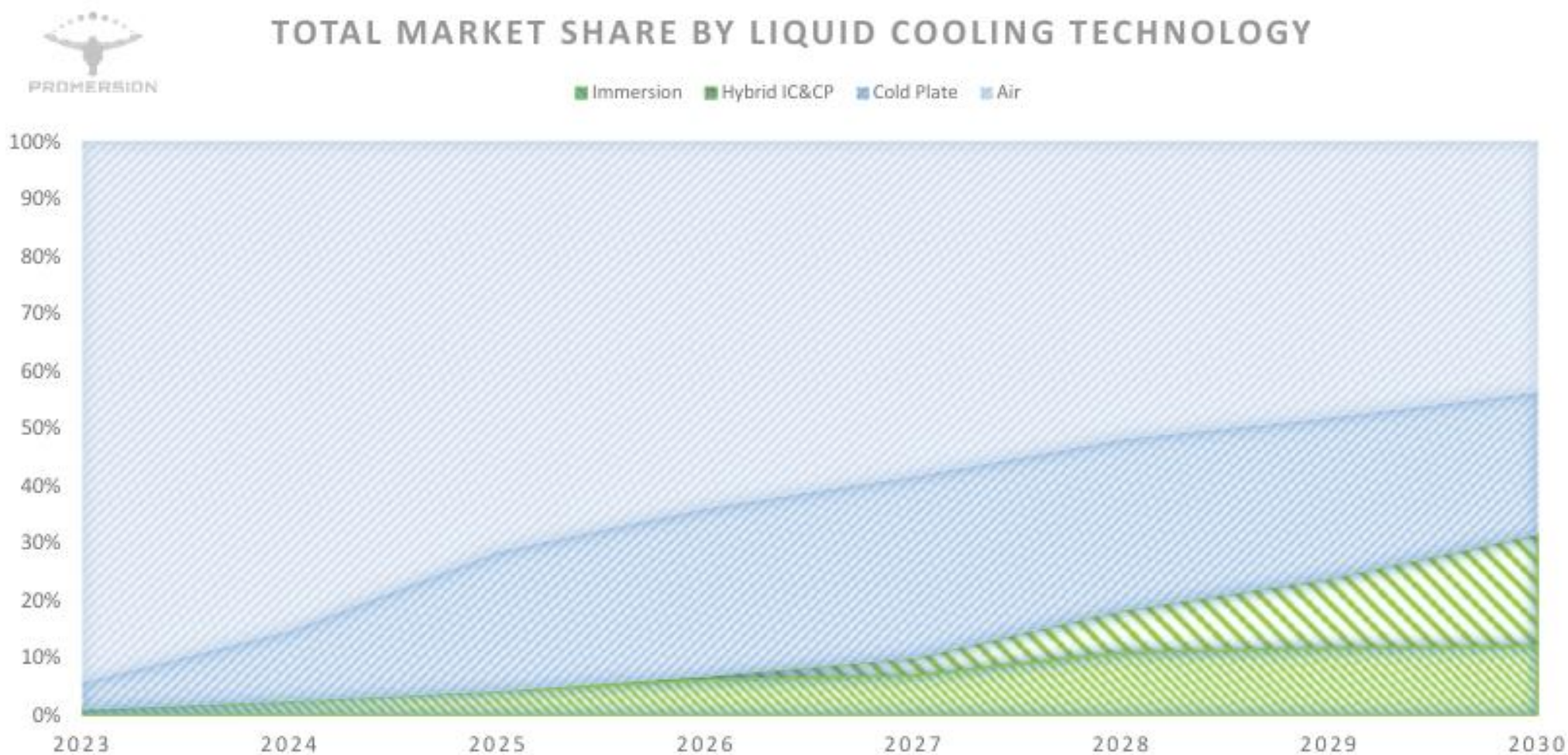
- 2025 年第二季至第三季成長放緩（從 20% 降至 3%）
- 受冷板普及的影響，新系統預先安裝了冷板，成長影響很可能是暫時的，預計 2026 年將再次加速
- 2028 年的拐點分析得到了更廣泛的認可
- OEM 廠商對浸沒式散熱技術的參與度大幅提高
- 更多策略性的超大規模試點/研究

五年預測

- 市場正朝著高度多元化的 IT 設備格局發展
- 強調 2024 年的轉捩點（冷板式）和 2028 年（浸沒式）
- 冷板式市場將在 2030 年達到飽和，因為目標市場將趨於飽和
- 預估 2024-2030 年複合年增長率分別為：冷板式 **47%**，浸沒式 **71%**



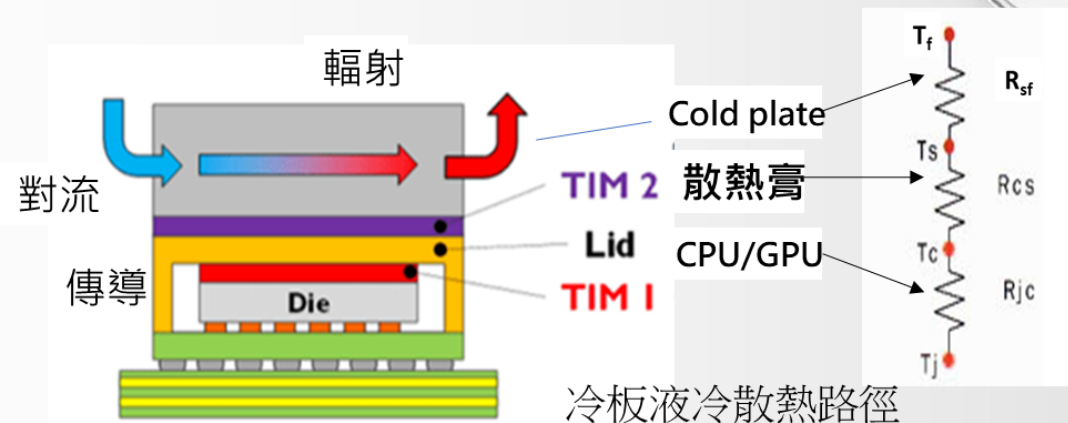
液冷技術市佔率



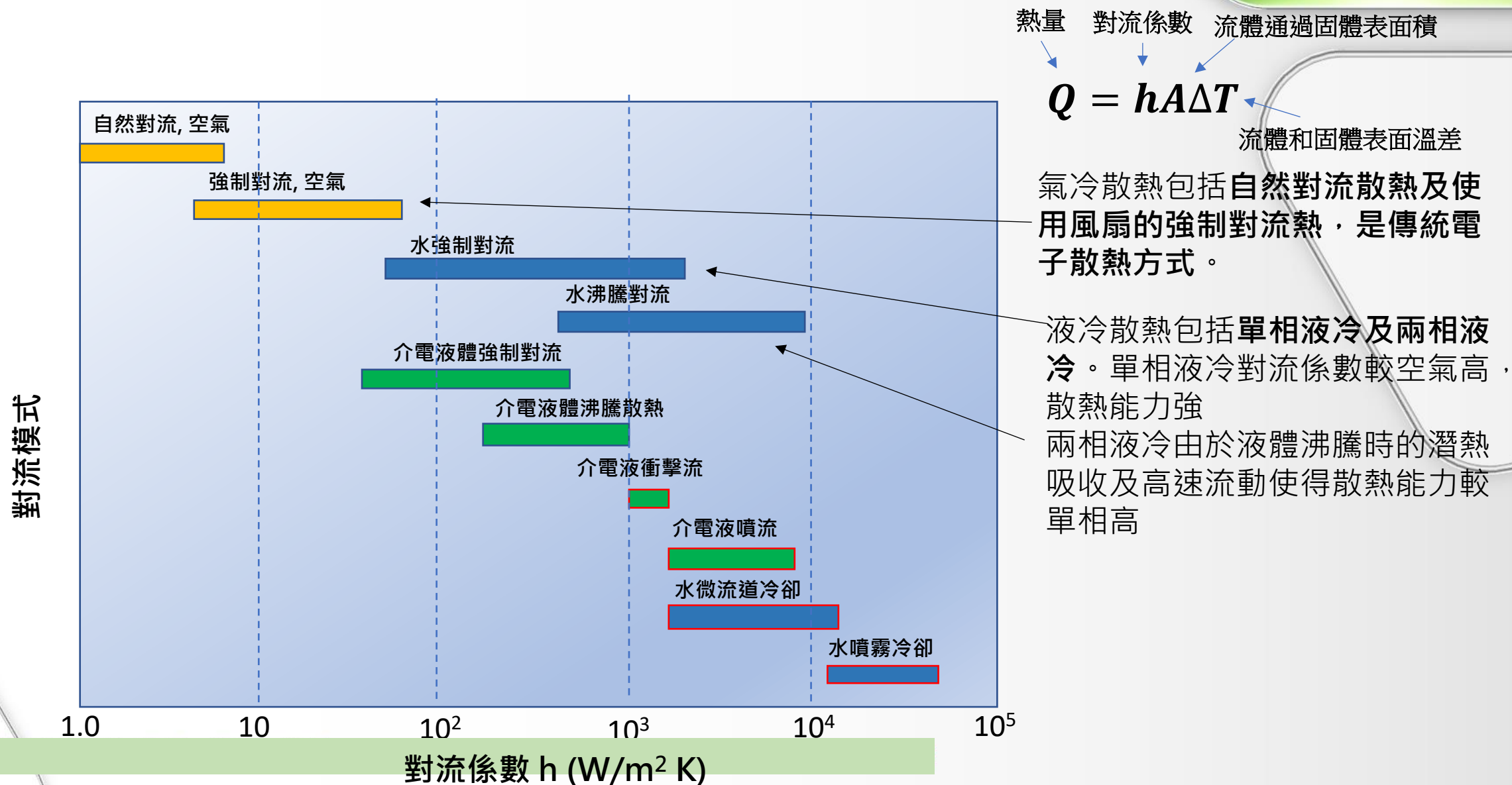
-
- Diagram illustrating the thermal management design process:
- Electronics Cooling Designers** (Left side): Focus on the internal components of the system, including the server chassis, the printed circuit board (PCB), and the microchip.
 - Facility Designers** (Right side): Focus on the external cooling infrastructure, represented by the server racks and the cooling system.
 - Heat Generation**: A blue arrow points to a detailed view of the microchip, with the text "熱在晶片的電晶體及互連間產生" (Heat is generated in the transistors and interconnects of the chip).
 - Heat Dissipation**: A blue arrow points from the cooling infrastructure to the text "熱逸散至環境中" (Heat is dissipated to the environment).

$$R_{th} = \frac{(T_j - T_f)}{P}$$

T_j : 晶片溫度
 T_f : 環境溫度(流體, 空氣)
 P : 發熱量

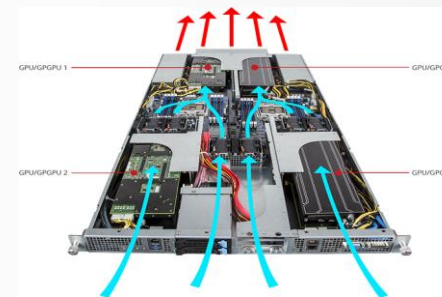
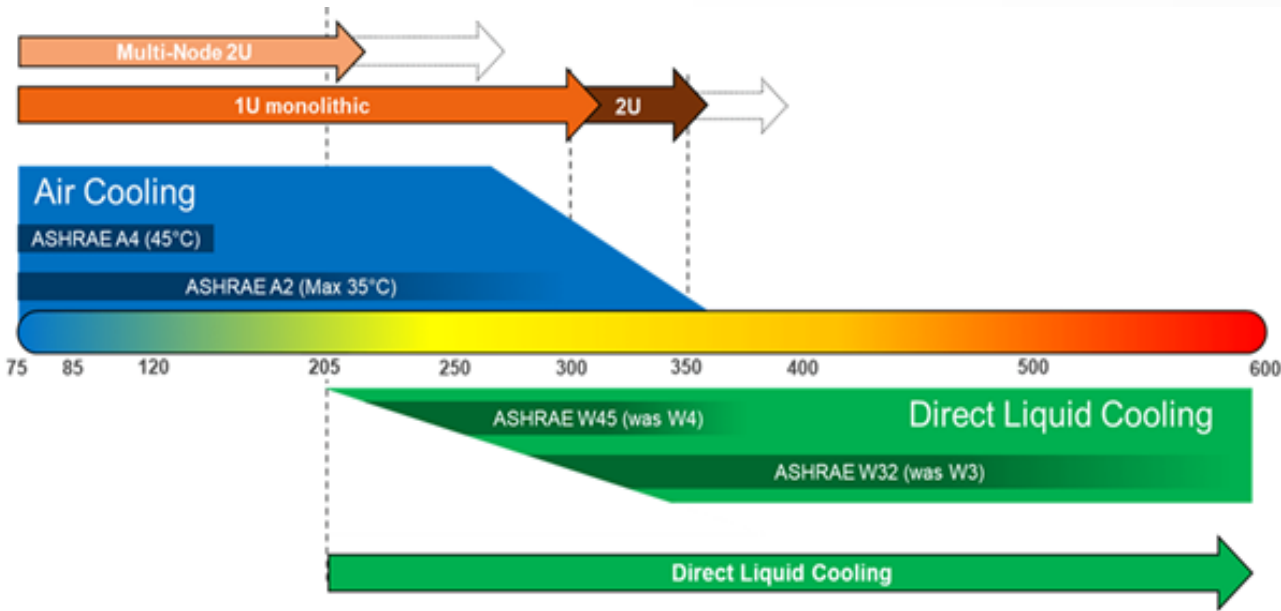


對流散熱模式

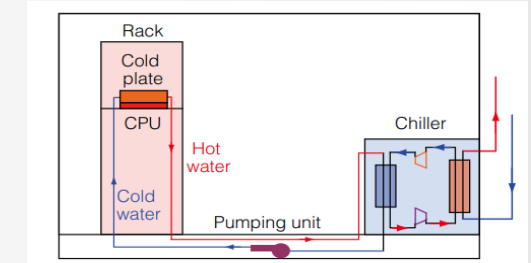
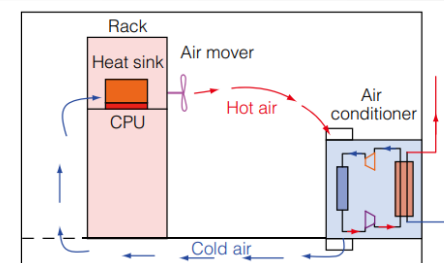


伺服器氣冷及液冷散熱

- 隨著功率從 300 W 走向 400 W，從風冷的角度來看，標準 1U 和 2U 伺服器變得更加困難。
- 採用液體冷卻，則可以輕鬆在 W45 (入口溫度最高 45°C) 下運作。隨著晶片功率的進一步增加，預計設施水溫可能必須降低。CPU及GPU轉向更大的封裝尺寸可能有助於滿足未來的冷卻需求。



(a) 伺服器氣冷散熱 (b) 伺服器液冷散熱



伺服器機櫃 (a) 氣冷散熱 (b) 液冷散熱架構

氣冷和液冷效能比較

- 氣冷效能較液冷低，建置及維護成本低
- 液冷PUE較氣冷低，能源轉換效率高
- 台灣散熱產業投入液冷及氣冷產業程度高

$$PUE = \frac{\text{設施總用電量}}{\text{IT設備用電量}}$$

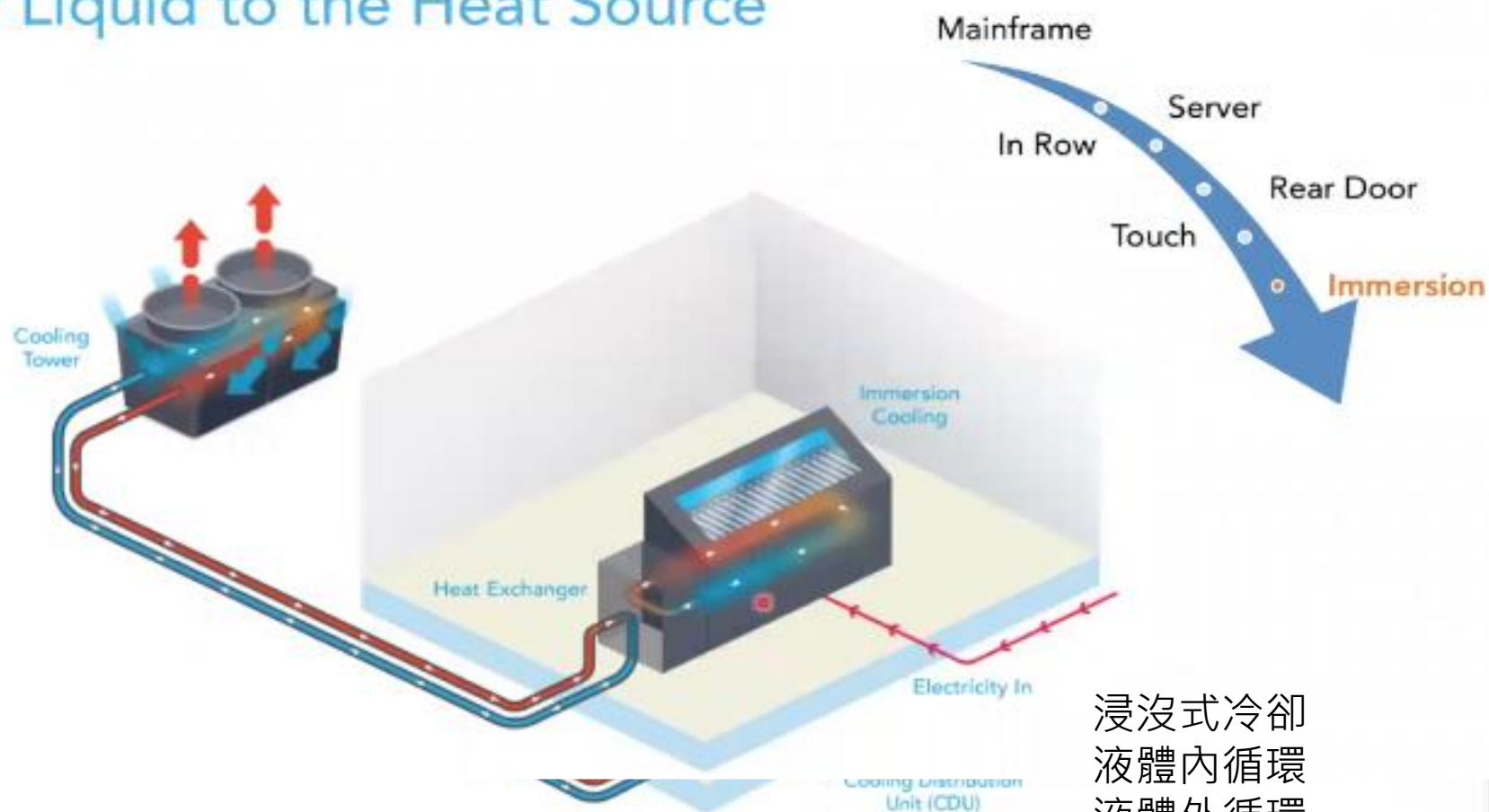
	氣冷	直接晶片冷卻	浸沒式冷卻
散熱能力	★	★★(★)	★★★★
建置成本	★★★★	★★	★
維護難易度	★★★★	★★	★
PUE	≥ 1.5	1.1~1.05	1.09~1.03

散熱方式	優 勢	布局台廠
散熱背板	對既有規格影響最小、可安裝在既有機種	雙鴻、奇鋐、台達電、雲達
直接式液冷	伺服器規格可適用、範圍較廣、降低風扇速度、噪音	超眾、奇鋐、建準、泰碩、高力、台達電、緯穎、和碩、英業達、雲達、技嘉、華碩
沉浸式液冷	最具有散熱效率、最小化風扇速度、靜音	奇鋐、建準、高力、台達電、緯穎、鴻佰、英業達、技嘉、神雲、華碩、和碩

資料來源：資策會MIC 製表：侯冠州

伺服器設施由氣冷到液冷

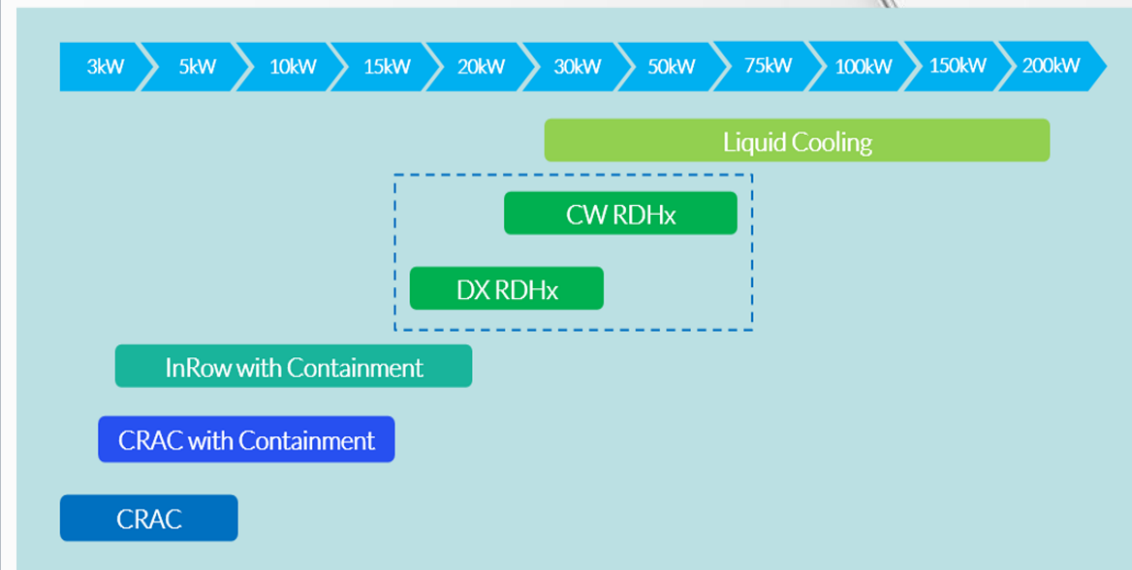
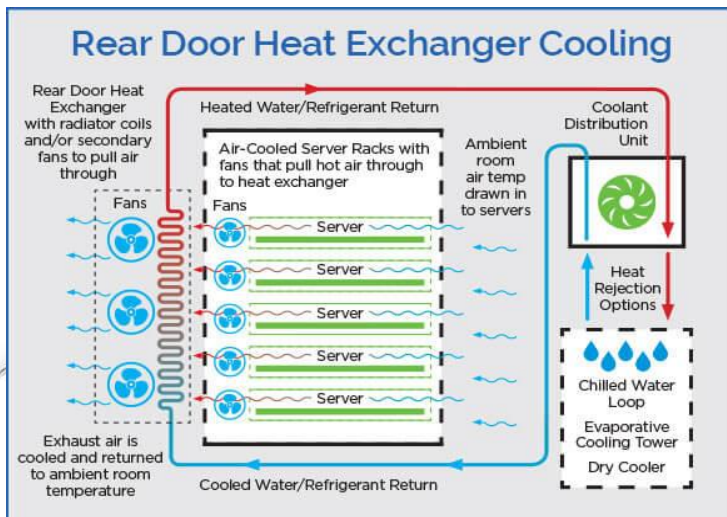
Proximity of Liquid to the Heat Source



浸沒式冷卻
液體內循環
液體外循環

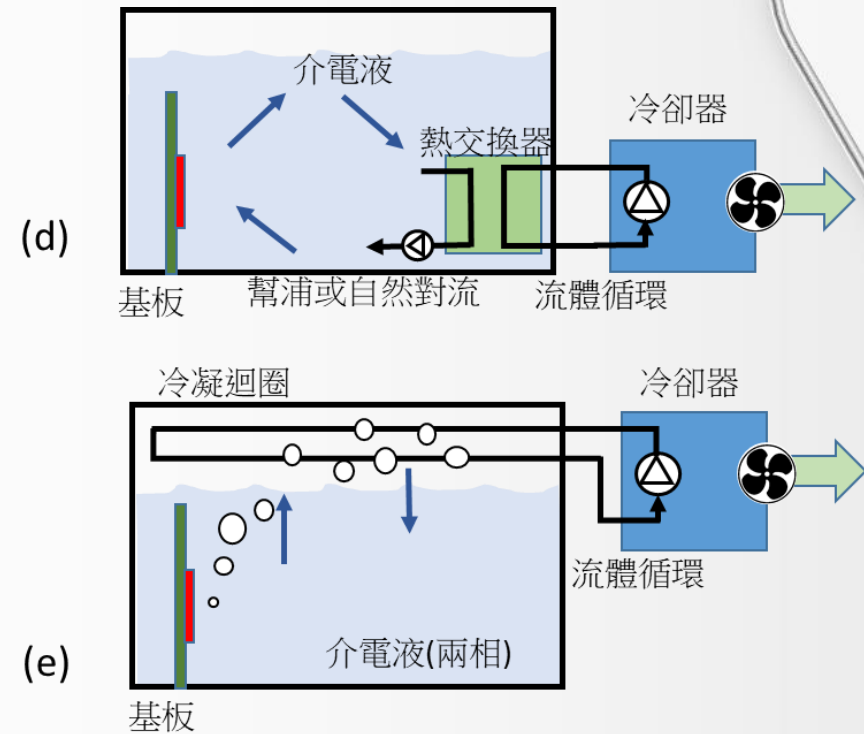
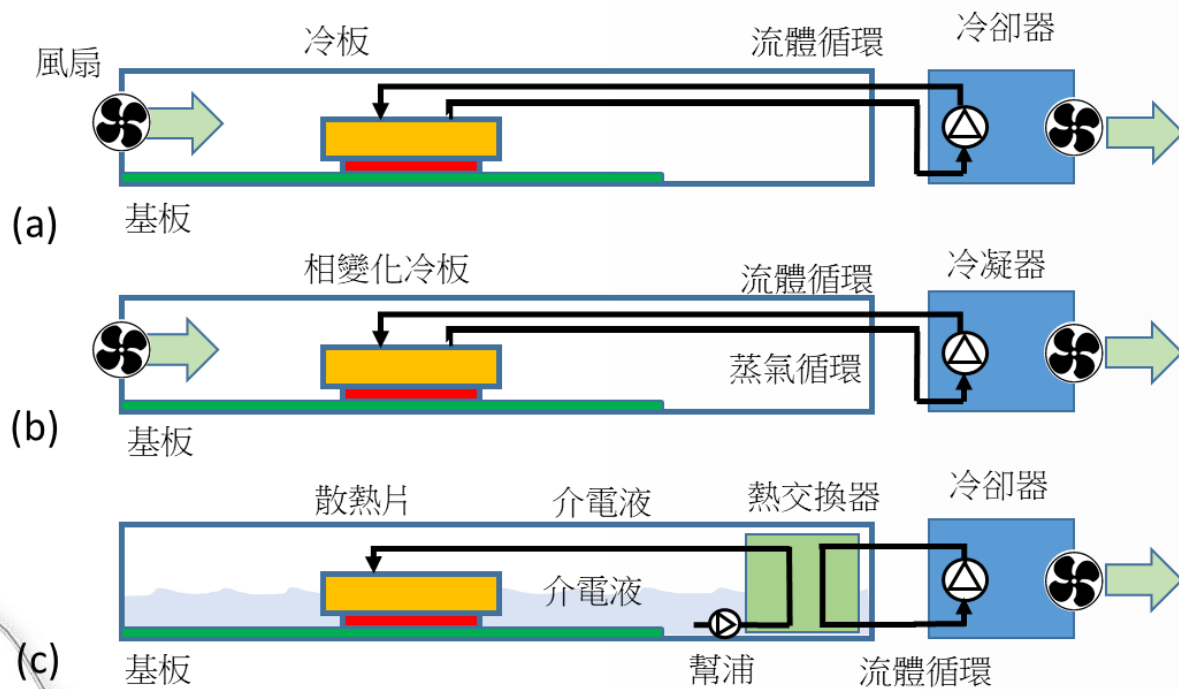
後門熱交換器

- 後門熱交換器液冷可將每台機櫃的最大負載提升至30~50KW，所以也有助於提升機櫃使用率，節省機房空間。
- 冷水式後門熱交換器利用冷卻水通過熱交換器
- 直接膨脹式後門熱交換器利用冷媒冷卻熱交換器



不同液冷形式

- 液冷散熱由導熱結構冷卻方式可分成直接晶片冷卻及浸沒式冷卻。
- 由流體對流形式可分成單相及雙相散熱。



(a)直接至晶片冷卻單相液冷 (b)直接至晶片冷卻雙相液冷 (c)浸沒式機殼型式單相及雙相液冷
(d)浸沒式管型式單相液冷 (e)浸沒式管型式雙相液冷

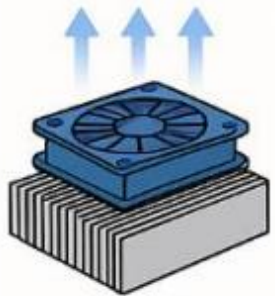
冷板散熱工作流體

- 主要冷卻液包括水、丙二醇(PG25, PG55)及介電液
- 所使用的元件材料(金屬, 塑膠, 橡膠等)需滿足相容性測試
- 流體狀況監測PH, 醇濃度, 金屬含量, 及濁度等
- OCP會議針對水質檢測提出討論

	Treated Water	Inhibited Propylene Glycol (PG 25, PG 55)	Dielectric Fluid (Two-Phase)
Thermal Performance	High	Medium	High (when at saturation)
Maintenance Schedule	High	Low	Low
Risk of Corrosion	Low	Low	None
Risk of Biofouling	Medium	Low	None
Electrical Conductivity	High	High	Very Low
Environmental Concerns	Low	Glycol discharge	PFAS, ODP, GWP
Operating Pressure	Low	Low	Low or High
Relative Cost Scale	\$	\$	\$\$\$

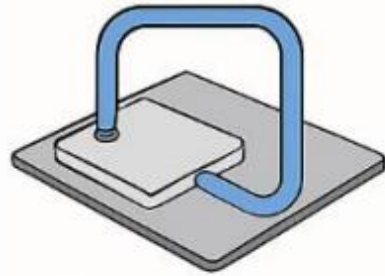
單相式冷板及兩相式冷板

Air Cooling



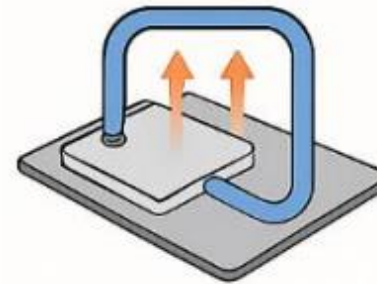
- 傳統風扇與散熱器
- 高耗能
- 高密度系統的散熱能力有限

Single-Phase
DLC(SPDLC)



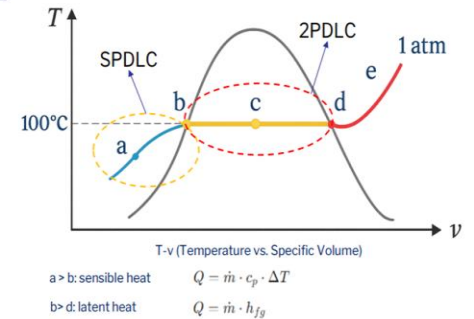
- 冷卻劑保持液態
- 在冷板間循環
- 比空氣冷卻效率更高
- 用於高效能運算 (HPC) 和人工智慧 (AI) 工作負載。

Two-Phase
DLC(2PDLC)



- 冷卻劑遇熱沸騰並汽化
- 汽化潛熱帶走更多能量
- 泵送能耗降低，傳熱效率提高
- 通常使用工程流體，例如 3M Novec 或 Chemours SF33

Physics Principle



單相液冷冷板

- 單相直接液冷 (SPDLC)
 - 內部設計特點：
 - 微通道或針狀翅片：在銅或鋁基底上蝕刻或加工出狹窄的液體通道（寬度約 0.2–1 mm）。增加傳熱表面積。
 - 流路：入口 → 通道/翅片 → 出口（冷卻液保持液態）。
 - 集流室：有助於在冷卻液進入微通道之前將其均勻分配。
- 利用切削等製程可以將流道間距降至**0.08mm**，熱阻降至**0.02°C/W**以下

$$Q = \dot{m} \cdot c_p \cdot \Delta T$$

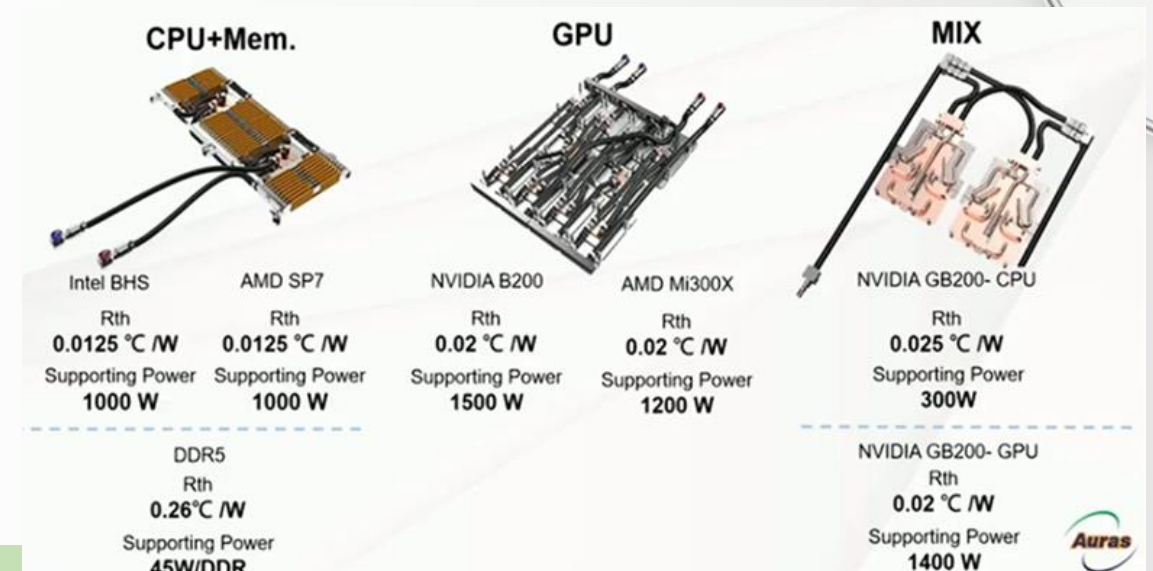
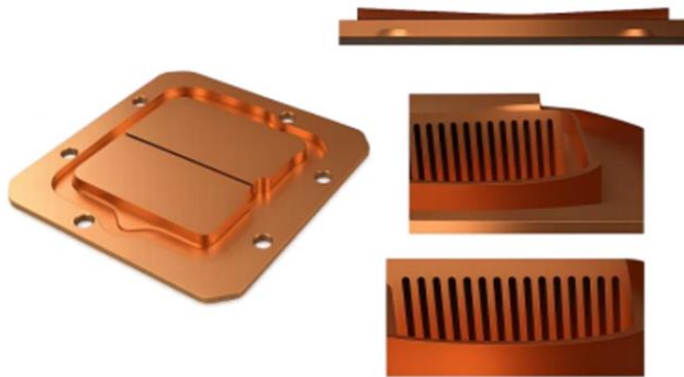
Sensible Heat Equation

Q: total heat absorbed by the coolant

M: mass flow rate of coolant (kg/s)

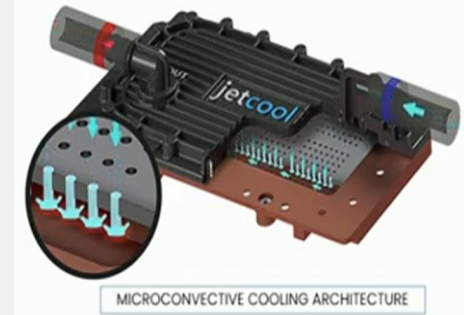
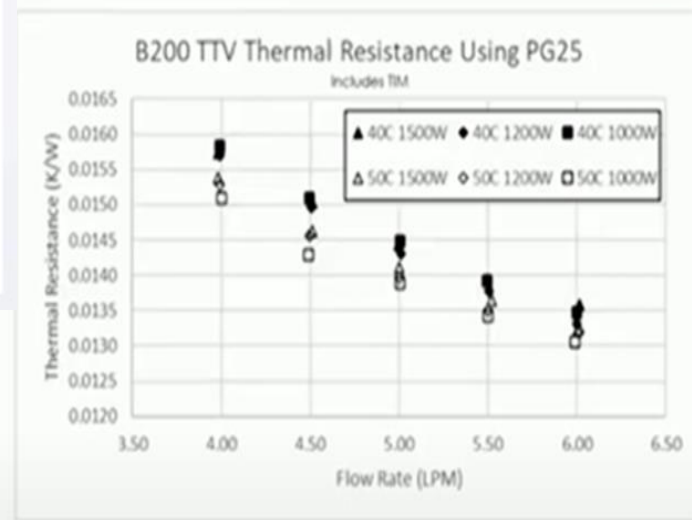
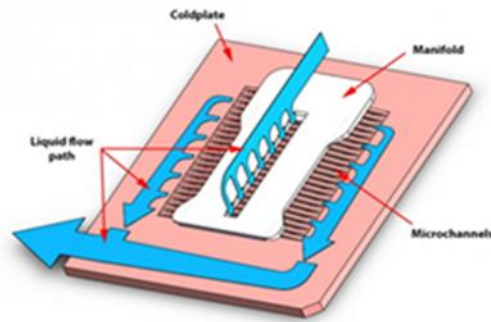
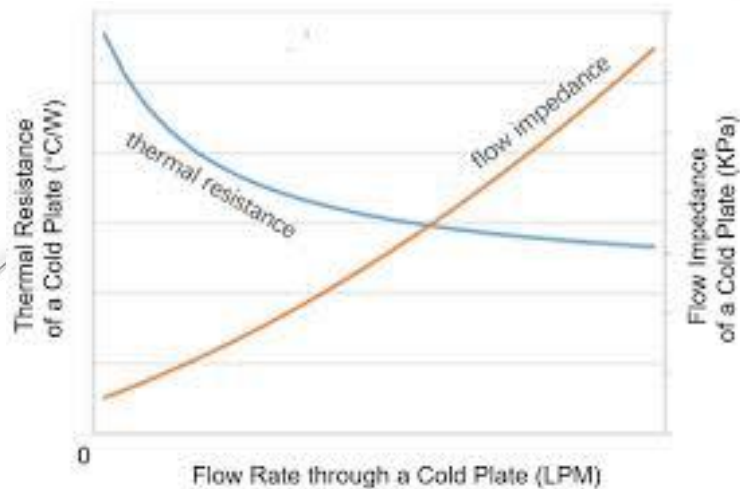
Cp: specific heat capacity of the coolant (J/kg·K)

ΔT: temperature rise of the coolant (K)



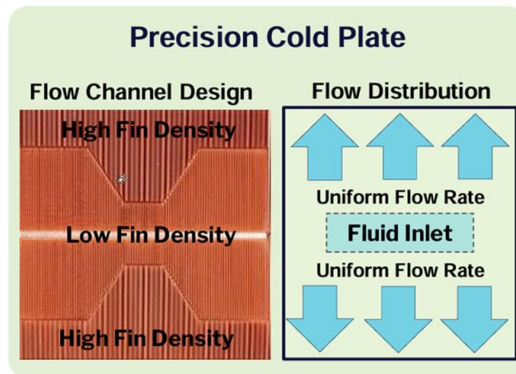
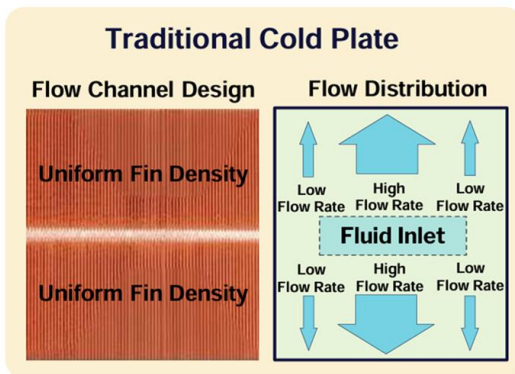
單相冷板結構

- 單相冷板熱阻值隨流量增加而降低，但壓降隨流量增加而增加。
- 目前利用切削等製程可以將流道間距降至0.1mm，熱阻降至0.02°C/W。
- 採用噴流及衝擊流可較傳統平行流冷板散熱效果更好。



精密冷板技術

Precision Flow Channel Design Boosts Cooling Efficiency



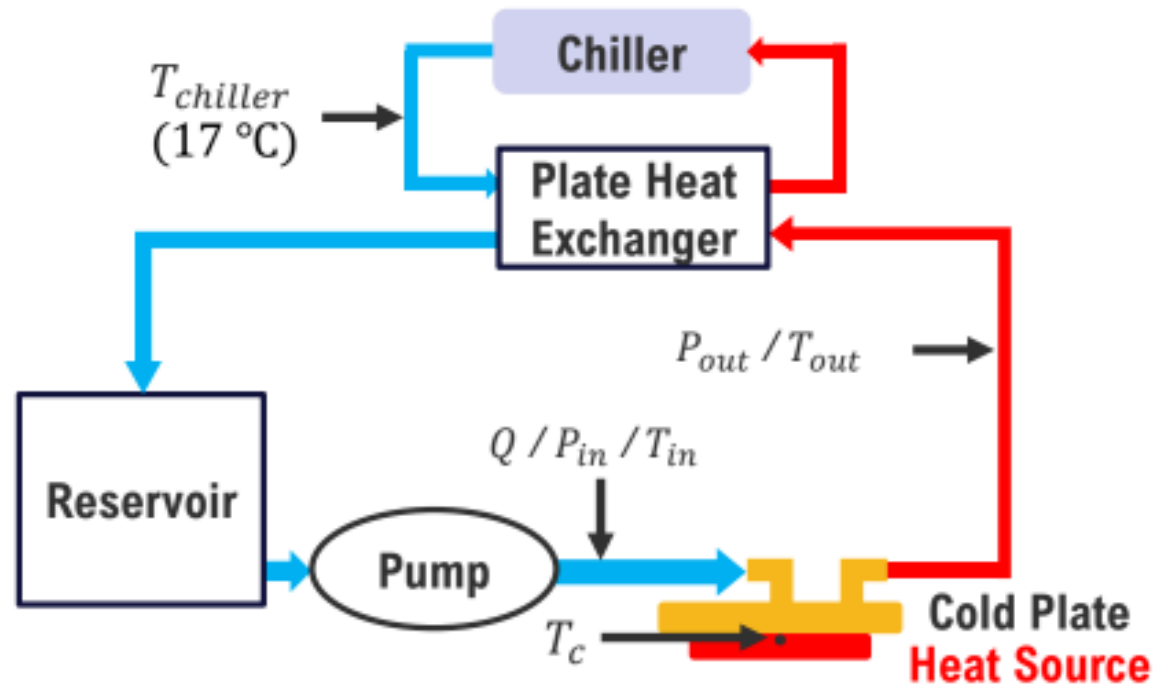
Precision Cold Plate Improved 18.2% Thermal Resistance

Under 2.0 LPM/kW, the precision cold plate has better thermal performance in the main die and HBM

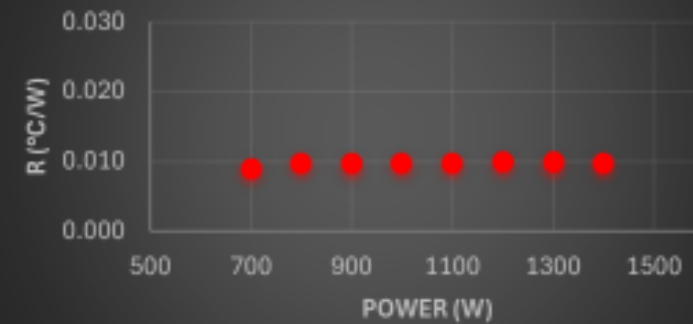
	Traditional Cold Plate	Precision Cold Plate
LPM/kW	2.0	2.0
Pressure drop (psi)	5.3	8.9
Main Die Ave. Case Temperature (°C)	80.4	73.3 (↓7.1°C)
Main Die Ave. Thermal Resistance (°C/W)	0.022	0.018 (↓18.2%)
HBM Ave. Case Temperature (°C)	60.6	49.6 (↓11.0°C)

1KW級冷板

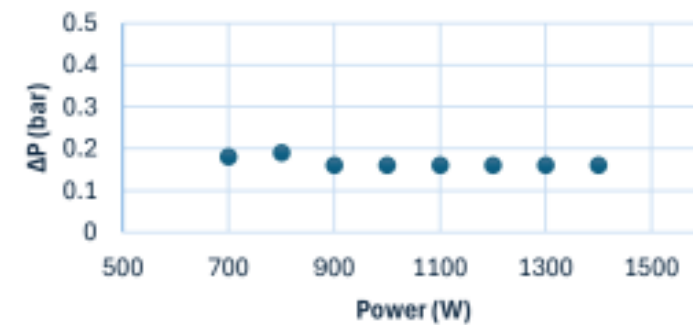
Cold Plate Test
Pump Flow Rate = 1 LPM
Coolant: PG25



Single Phase Cold Plate



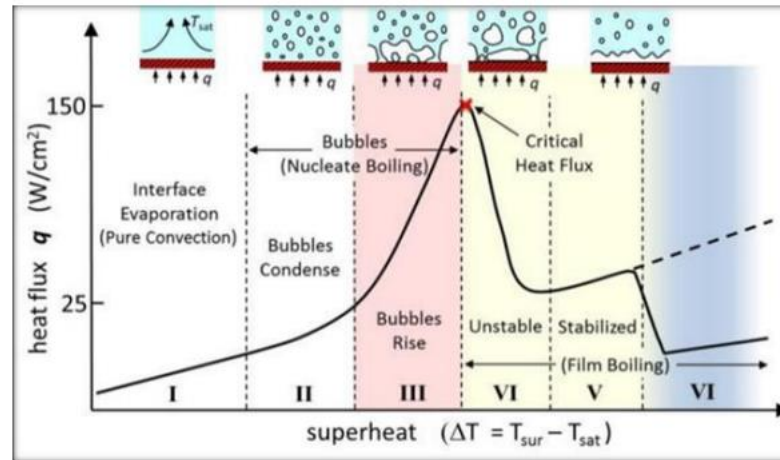
AMD SP7 (PG25)



兩相冷板

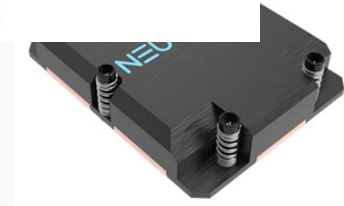
- 兩相直接液冷 (2PDLC)
- 內部設計特點：
 - 沸騰表面/芯吸結構：
 - 多孔燒結金屬層或微圖案芯吸透過捕獲蒸汽氣泡和控制液體蒸騰。
- 蒸汽逸出路徑：
 - 沸騰產生的蒸汽通過蒸汽出口體入口分開。
- 液體分佈層：
 - 將流入的液體均勻分佈在沸騰表面上。
- 毛細管或重力回流（可選）：
 - 部分板包含延伸至回流路徑的芯吸層，用於回收冷凝液。

Figure 3. Schematic of Pumped 2 Phase (P2P) Direct Liquid Cooling (DLC) technology

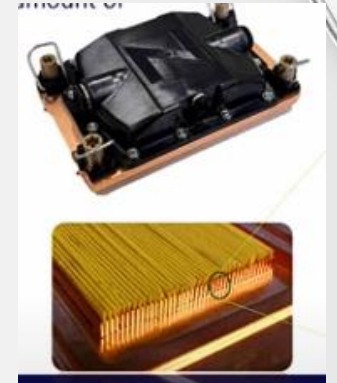


Q : heat removed
 m : mass flow rate of vapor
 h_{fg} : latent heat of vaporization (J/kg)

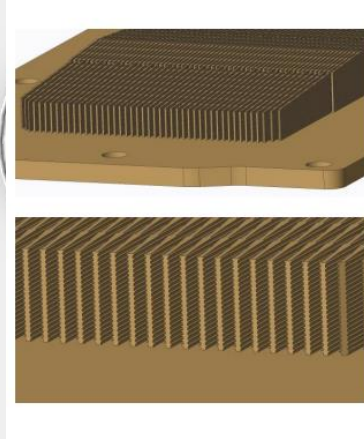
$$Q = \dot{m} \cdot h_{fg}$$



Accelsius

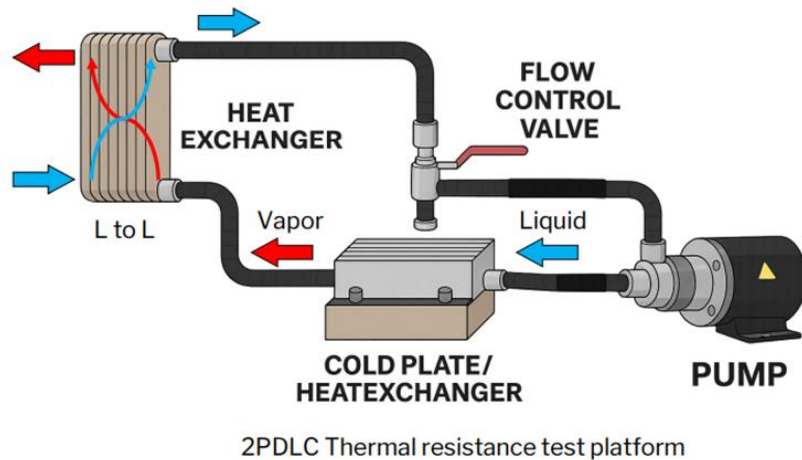
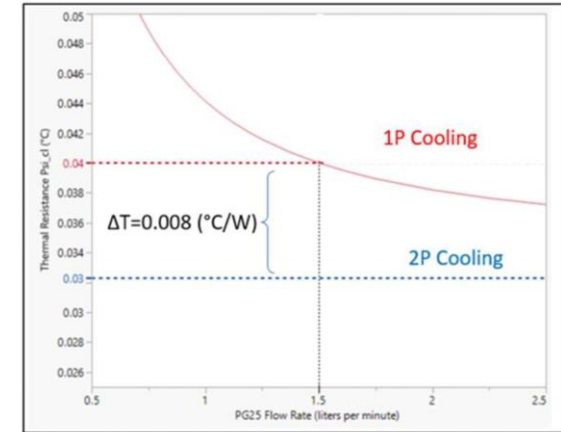


Zutacore



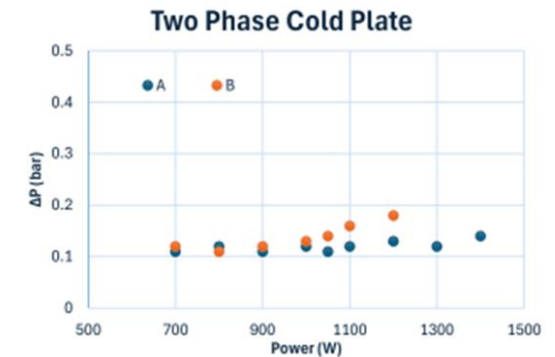
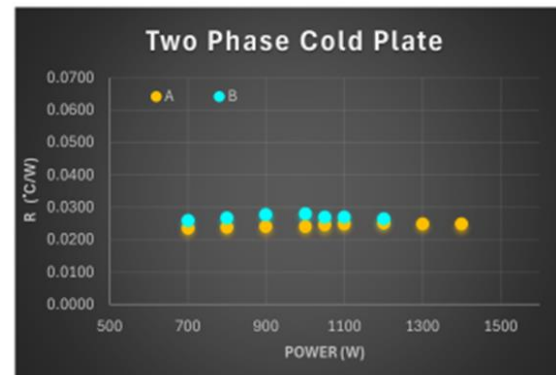
兩相冷板

- 液體進入冷板後蒸發為蒸氣，由熱交換器將蒸氣散熱轉為液體。
- 雙相較單相冷板散熱效能提升約15%~20%。
- 隨著流量增加散熱效能增加，可達 $250\text{W}/\text{cm}^2$ 。

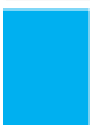


Cold Plate Test
Pump Flow Rate = 1 LPM

Coolant	Fluorinated liquid A	Fluorinated liquid B
Boiling Point (°C)	56	47
Specific Heat (J/kg·°C)	1100	1279
Latent Heat (J/g)	88	93



單相及雙相液冷比較

	Pumped Two-Phase Cooling	DLC	Air Cooling
Heat Flux	 ~4x	 ~2x	 ~1x
Rack Power Density	 ~4.4x	 ~2.8x	 ~1x
Pumped Power	 Saving 65%	 ~ 1x	N/A

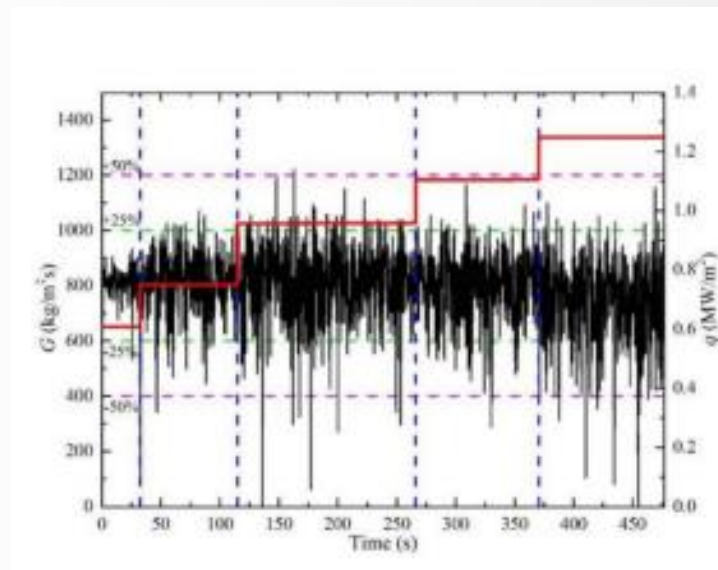
PROS & CONS

	Pumped Two-Phase Cooling	DLC	Air Cooling
	Refrigerant	PG25	Air
	• Low OPEX • For hyper heat flux • High energy efficiency	• Lower OPEX • For high heat flux • High energy efficiency	• Low CAPEX • Easy maintenance
	• High CAPEX • Higher operating pressure • GWP/PFAS	• Risk of leakage • Higher flow rate	• High OPEX • Poor PUE

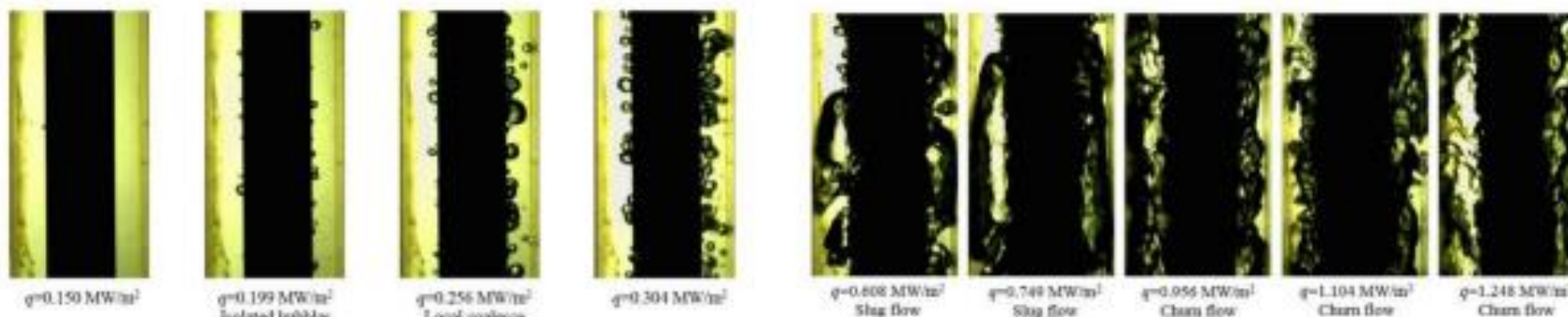
泵送式兩相冷卻的困難點

需要解決哪些挑戰？

- 不穩定性
 - 不同的加熱速率對應不同的流動模式
 - 不同的流動模式會導致不同的流量不穩定性
- 維護
 - 更嚴格的安全要求



Flow boiling patterns correspond to different heat fluxes



References: H. Liu, W. Liu, D. Chen, H. Liu, J. Qin, P. Yan, P. Li, Experimental investigation on flow oscillation of boiling two-phase flow in a single rod channel, Annals of Nuclear Energy 158 (2021) 108281.

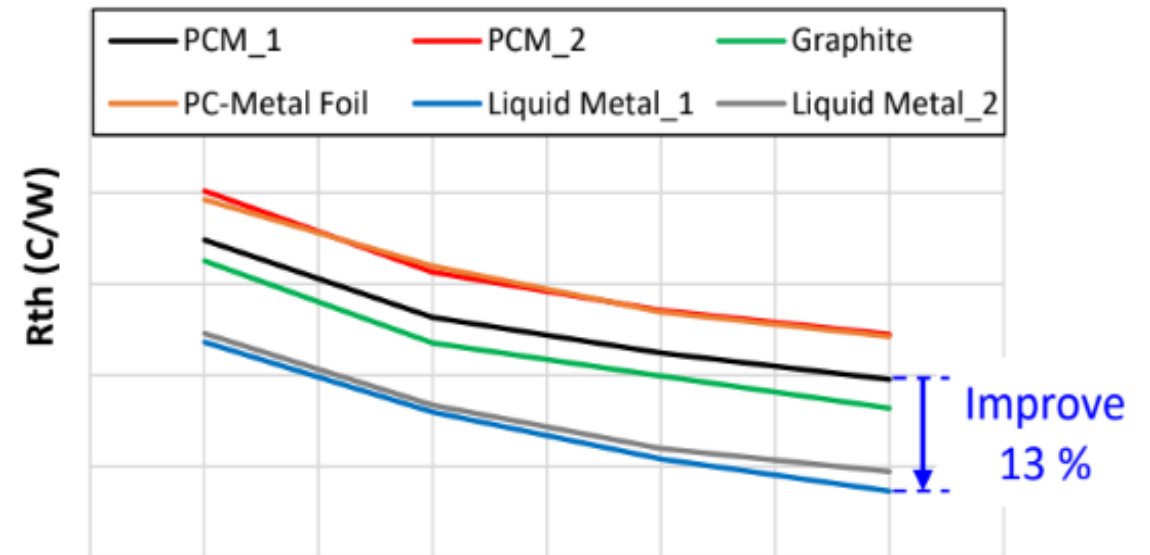
TIM材料比較

- Goal:
- Survey various types of Thermal Interface Materials (TIMs) and benchmark their performance against Phase Change Material (PCM)
- Testing Unit:
- Copper TTV (700W): Used for TIM survey
- Real GPU (700W): Used for quantify the TIM performance in a real chip environment
- Test Condition:
- PG25, Flowrate 1~1.6 LPM/GPU @30 °C coolant inlet temperature

TIM	PCM-1	PCM-2	Graphite	PC-Metal Foil (In+Bi+Sn)	Pure LMT-1 (Ga+In)	Pure LMT-2 (Ga+Sn+In)
K (W/m-K)	8.5	7.5	~42	18	*28	*26
Rank	4	5	3	6	1	2
Setup						

TIM材料比較

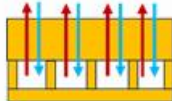
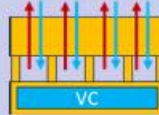
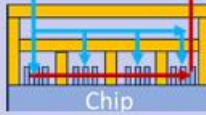
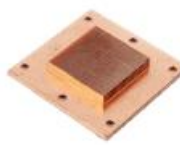
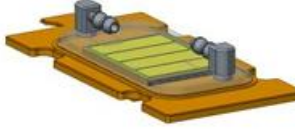
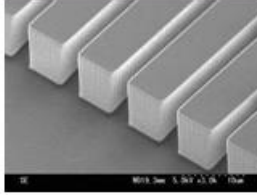
TIM	Rank of K-value	Rank on GPU	Variation
PCM-1	5	4	▲ 1
PCM-2	6	6	-
Graphite	1	3	▼ 2
PC-Metal Foil	4	5	▼ 1
Liquid Metal-1	2	1	▲ 1
Liquid Metal-2	3	2	▲ 1



先進散熱技術

Advanced Cooling Solution

- Transistor Density **1.8X**
- Power Consumption Reduction **25%**
- Computing speed promote **15%**

Cooling Solution	Single Phase 			Two Phase  			
Flow Rate	0.75PM~1.5LPM	1.0LPM~2.0LPM	2.0PM~3.0LPM	1.0LPM ~1.5LPM	1LPM~1.5LPM		
CDU output Pressure	200kPA	350kPA	500kPA	100kPA	100kPA		
Material							
	Pure Copper			MIX Copper (Graphene & diamond)	MIX Copper and Vapor Chamber	Silicon	
Cold Plate	  			  			
	Thickness:0.2mm Pitch: 0.35mm~0.4mm			Thickness:0.15mm Pitch: 0.3mm		Thickness:0.1mm Pitch:	
	1. Graphene Copper 2. Diamond Copper			Single Phase VC Cold Plate		2Phase VC Cold Plate	
Power consumption	700W	945W	1275W	1720W	2320W	3130W	4225W
Photolithography	3N	2N	A14	A10	A7	A5	A3
Year	2022	2024	2026	2028	2030	2032	2034

新散熱材料

Next Gen Cold Plate Solution Roadmap



Diamond Alloy-250

- Diamond particle size: 250 micrometer
- Thermal conductivity coefficient: 680-730 W/mk

Next Gen_Diamond Alloy-50

- Diamond particle size: 50 micrometer
- Thermal conductivity coefficient: 750-800 W/mk

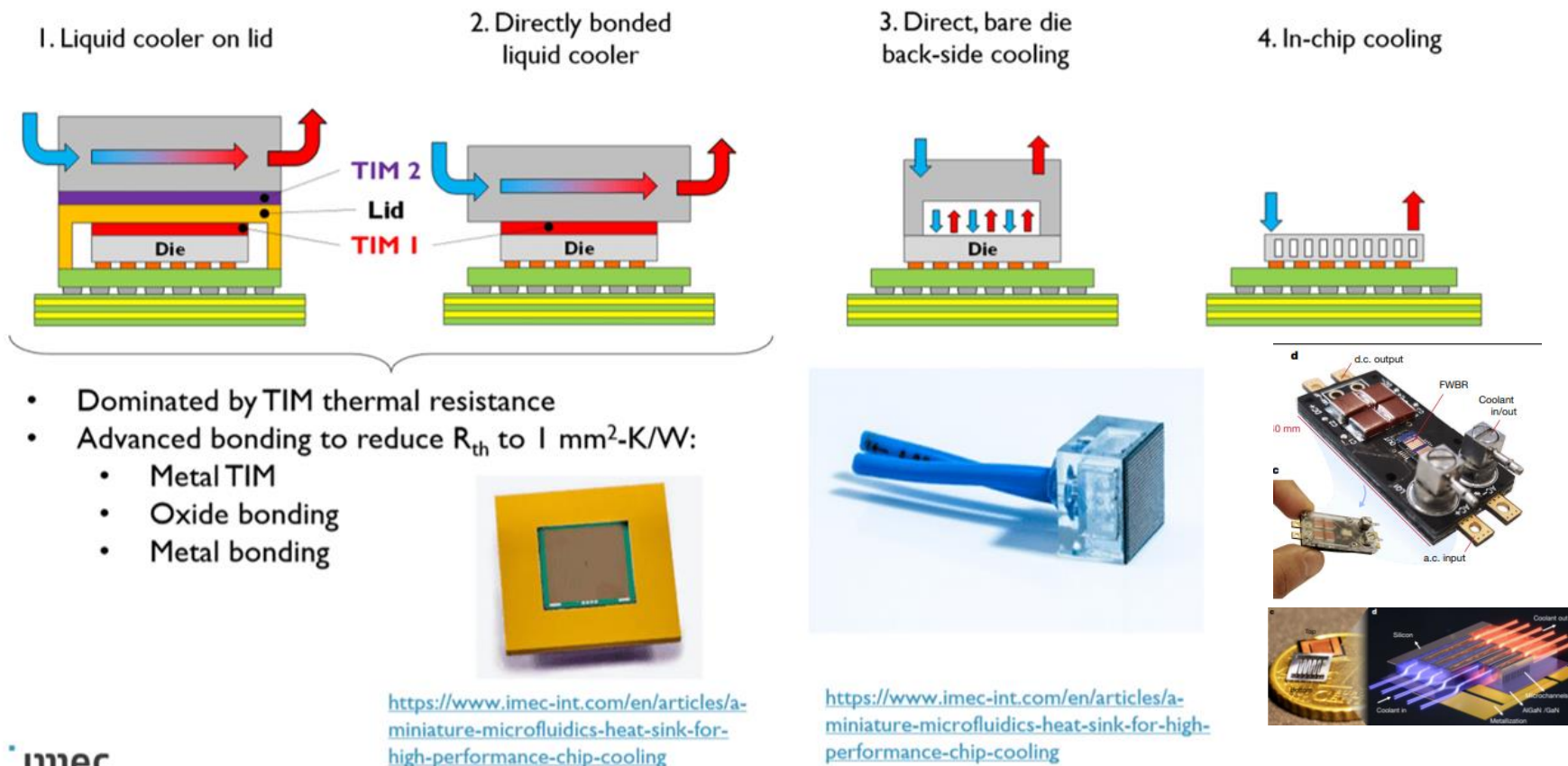
CTI 华测检测

测试样品 Test Sample	测试结果 Test Results
测试方向 Test Direction	纵向 Vertical
试验温度 Test temperature °C	25
厚度 Thickness mm	3.015
密度 Density g/cm ³	5.344
比热容 Specific Heat Capacity J/(g*°C)	0.439
热扩散系数 Thermal Diffusion Coefficient mm ² /s	311.666
导热系数 Thermal Conductivity Coefficient W/(m*K)	731.173

备注：本报告中的数据结果供科研、教学、企业内部质量控制、企业产品研发等目的用。
Remark: The testing data and result(s) in this report is(are) just for scientific research, education, internal quality control and product development etc.

結合元件及封裝設計

- 整合封裝將散熱路徑縮短以降低整體熱阻值



Microchannel Cooling

微流道散熱與晶片封裝結合

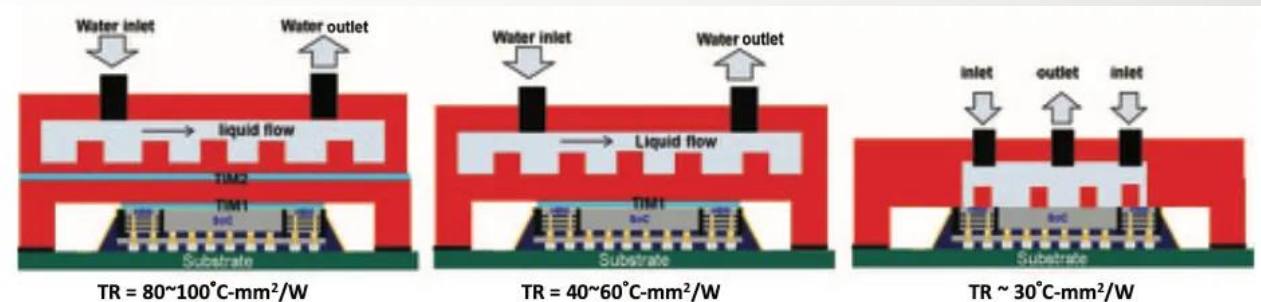
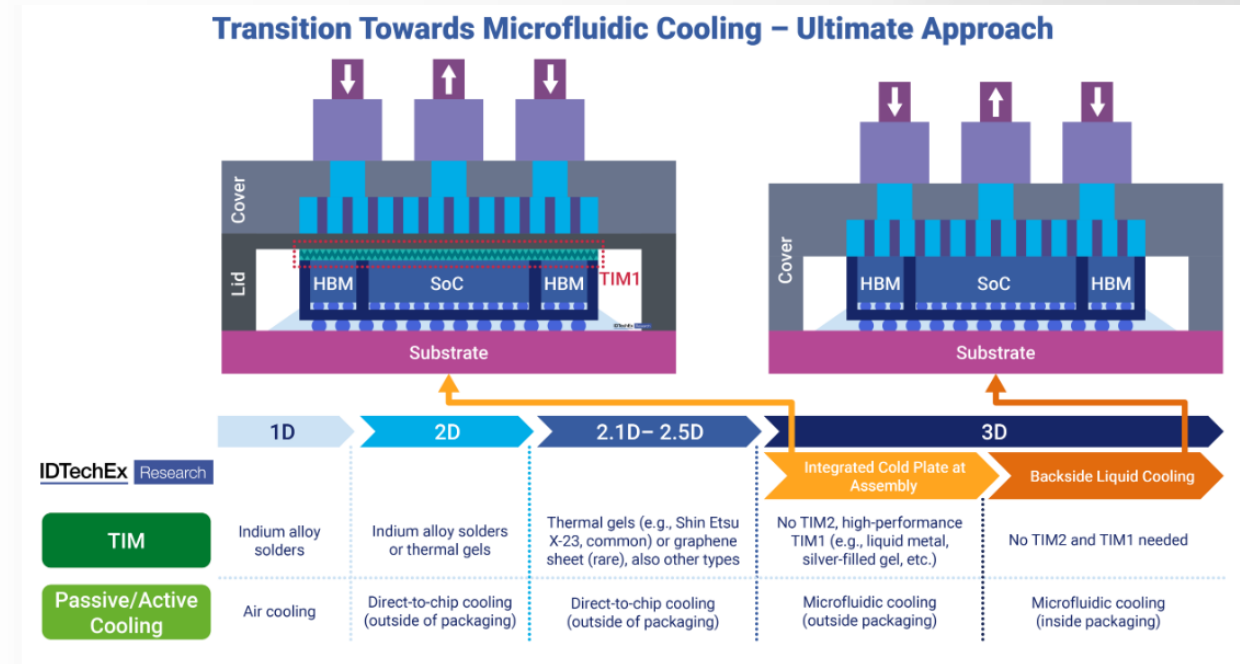


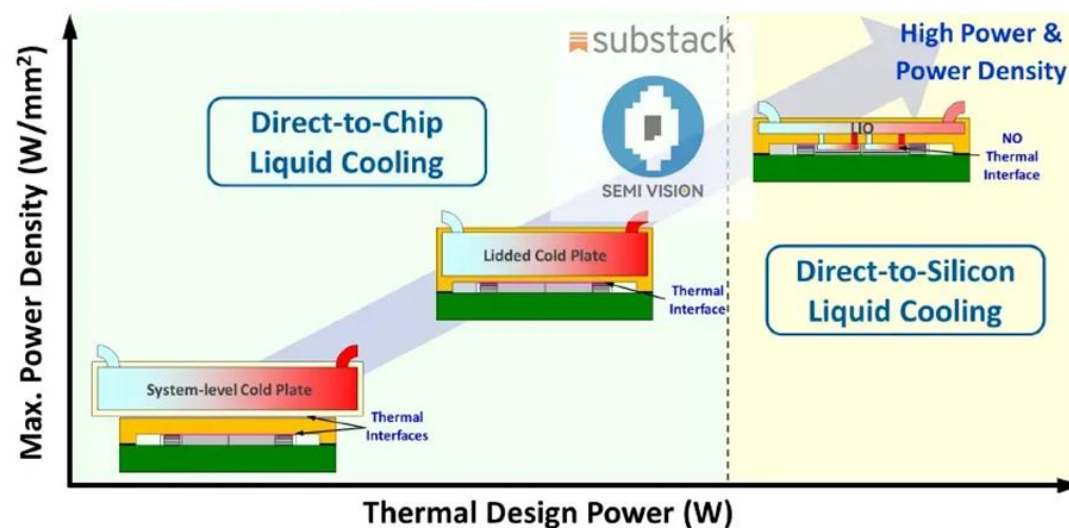
Fig. 1. 2.5D liquid cooling trends (source: TSMC)

Si Microchannel

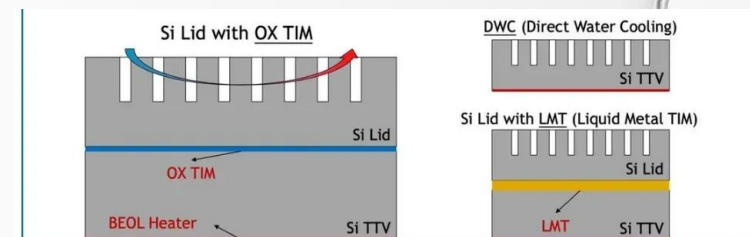
TSMC提出Si Microchannel散熱

2021

NO thermal interface in direct-to-silicon liquid cooling

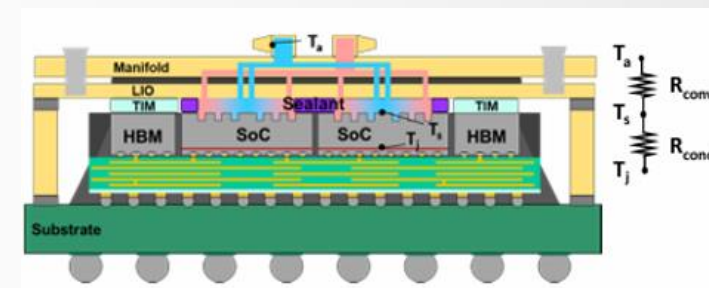


Yu-Jen Lien, Cheng-Chieh Hsieh, Terry Ku, Li Wang, Po-Ju Chen, Kuo-Chung Yee, and Douglas C.H. Yu
2024 IEEE 74th Electronic Components and Technology Conference | Denver, Colorado | May 28 – May 31, 2024



散熱瓦數2600W, 功率密度4.8 W/mm^2 .

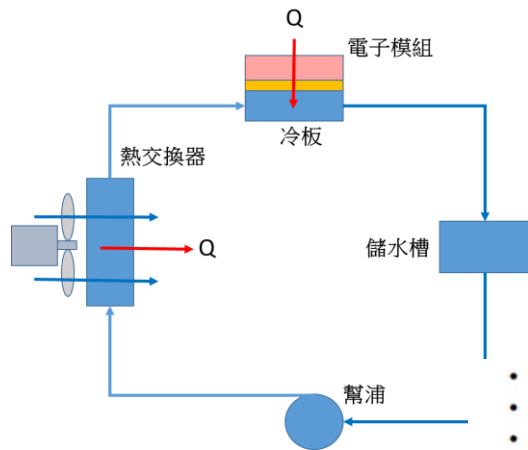
2025



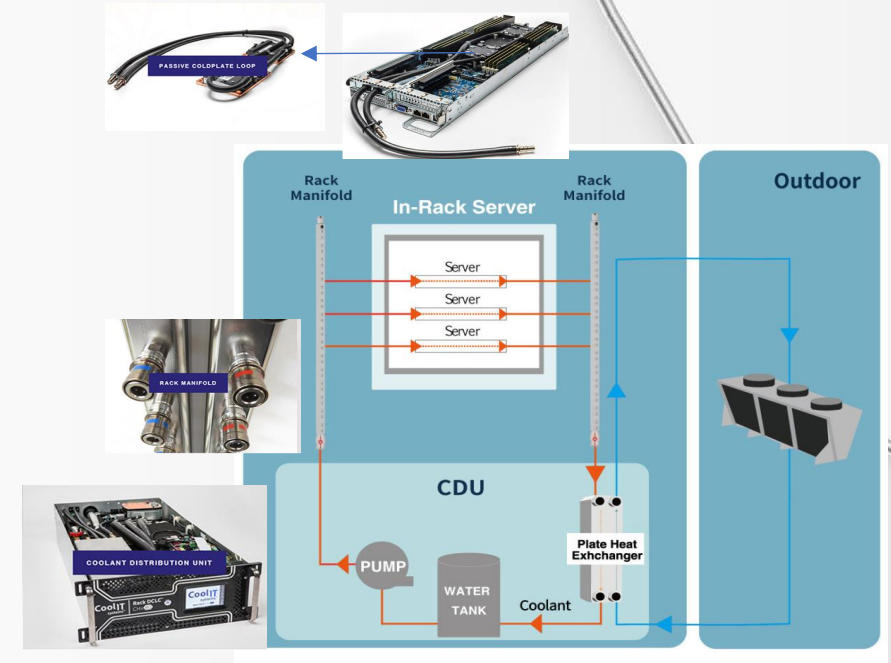
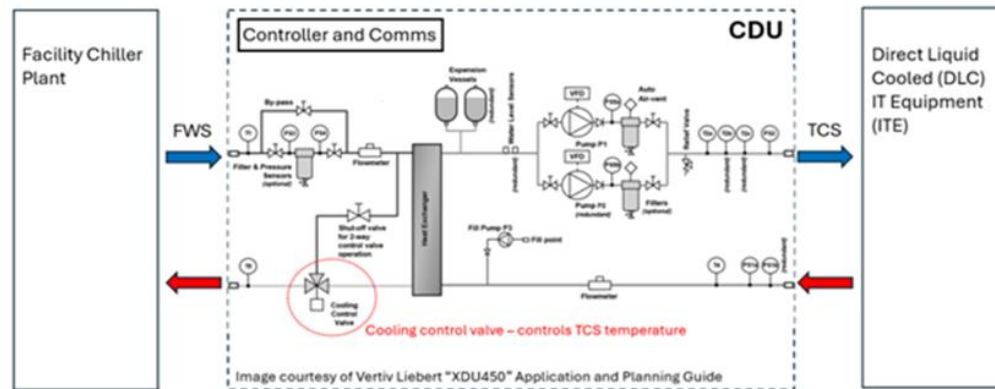
冷卻器能在 40°C 水流、流量 10 公升/分鐘時，耗散超過 3,000 瓦的均勻功率。功率密度2 W/mm^2

液體分布單元(CDU)

- 主機內經由液冷組件將大功率組件，如CPU、GPU、Memory、High-speed network card等發熱元件熱量帶走。
- 液冷系統架構包括冷板、熱交換器、管路及儲水槽等，實際應用上整合成為CDU。



- CDU provides isolation and control of the TCS cooling fluid
- Can be paralleled to reduce ATD and increase overall heat rejection capacity
- Can be configured as N+1, N+N, or standalone



CDU技術方向

降低阻抗

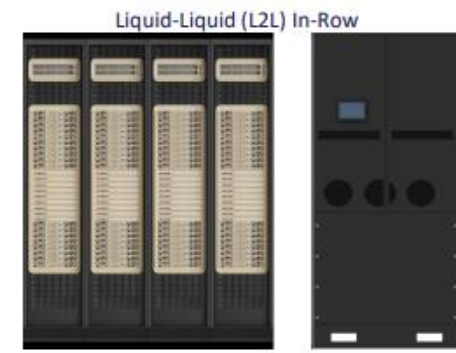
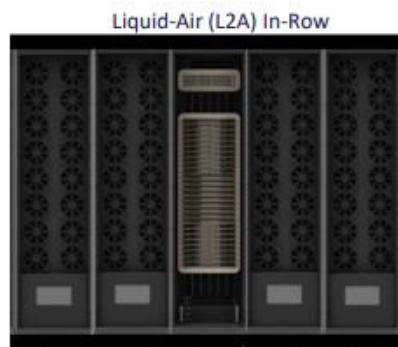
- 盡可能使用全通徑閘門/連接器
- 探索機架分線和快速連接 (QD) 的平行路徑

未來的擴充

- 根據未來流量決定管/軟管直徑
- 為額外的 CDU 和 ITE 機架分配分線和隔離閘
- 模組化解決方案
- 模組化的 CDU 和歧管設計可讓基礎設施隨負載擴展

• Typical Capacities

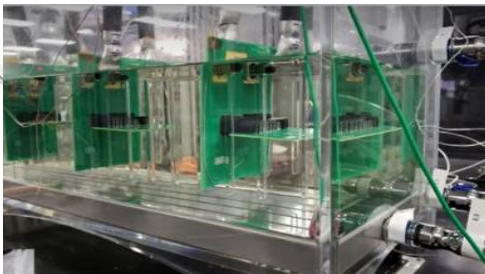
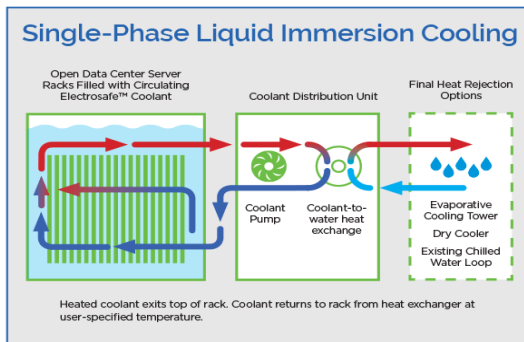
- L2A In-Row ~ 130 kW
- L2L In-Rack ~ 200 kW
- L2L In-Row ~ 1,600 kW



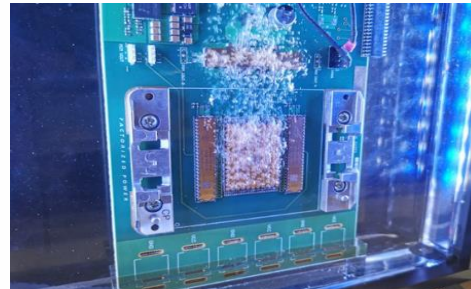
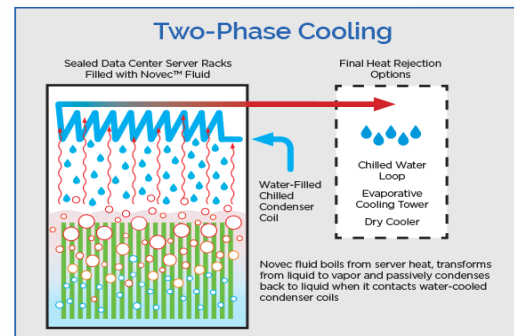
浸沒式冷卻比較

- 兩相式浸沒式冷卻散熱效能較單相高，但須採用沸點低的介電液，有補充問題及環保疑慮。
- 單相浸沒式冷卻介電液包括合成油及矽油等，兩相式浸沒式冷卻介電液則以碳氟化合物為主(FC-72、FC-77、HFE)。

單相浸沒式冷卻



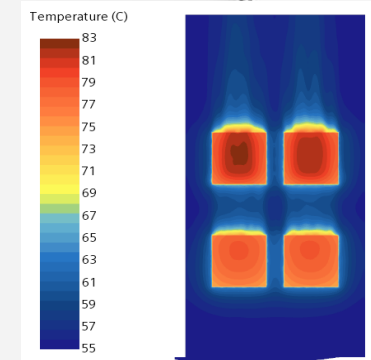
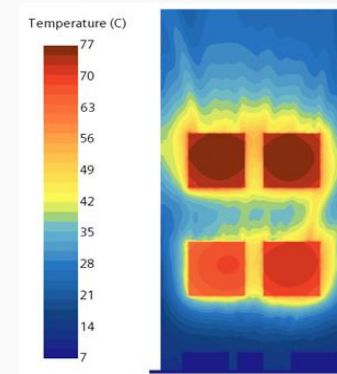
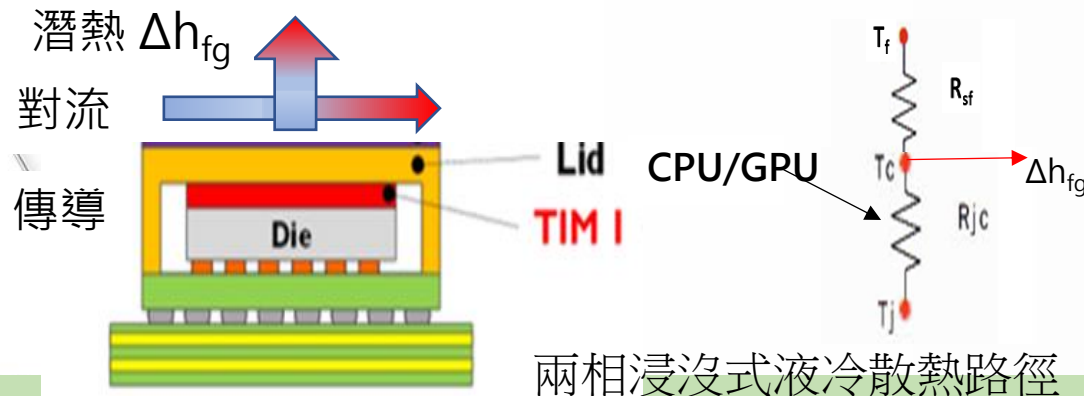
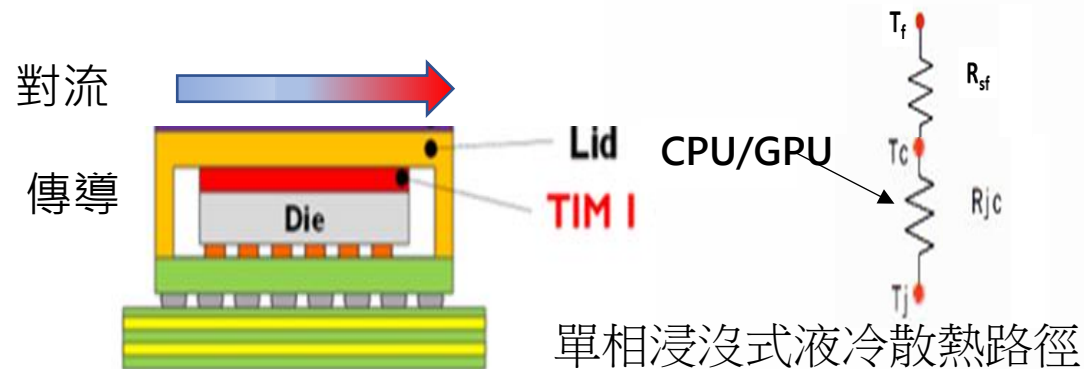
兩相浸沒式冷卻



	單相	雙相
散熱效能	★★★★☆	★★★★★
複雜性	★★★★★	★★★☆☆
效率	★★★★☆	★★★★★
流體損失	★★★★★	★★☆☆☆
操作成本	★★★★★	★★☆☆☆
環保	★★★★★	★☆☆☆☆
維護成本	★★★★★	★★☆☆☆

浸沒式冷卻技術比較

- 浸沒式冷卻散熱路徑短，熱阻低，兩相式浸沒式冷卻散熱效能較單相高。
- 單相浸沒式冷卻可朝提升接觸面積及流速設計，兩相式浸沒式冷卻可朝提升沸騰表面處理等設計提升散熱效能。

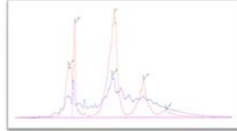
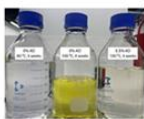
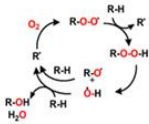


Cooling Type	Single phase immersion cooling	Two phase immersion cooling
Coolant	FC40	FC72
Power Dissipation(W)	100	150
Maximum Temperature (°C)	132@100W	83@150W
Power density(W/cm ²)	11.1	16.7

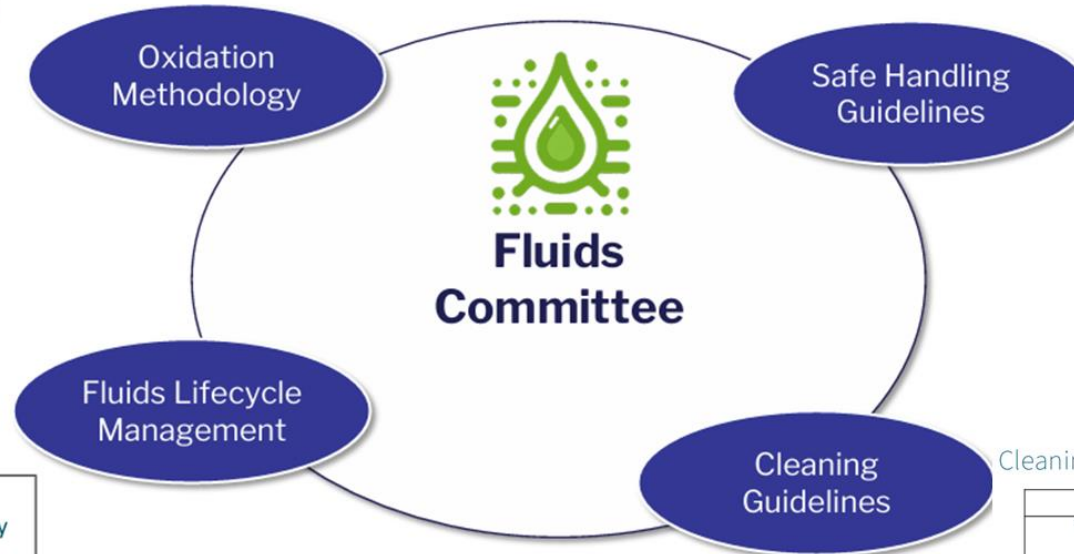
浸沒式冷卻技術流體需求

Problem statement

We currently have do not have a consistent way of quantifying oxidation in immersion fluids.



Vanja Boni



• Provides guidelines to best practices for the safe handling of:

- Hydrocarbons
- Esters
- Fluorinated liquids

Problem statement

How can I safely and sustainably dispose of or recycle my immersion fluid?



Destruction



Recycle

Cleaning Solvent Selection

Solvent	Advantages	Disadvantages
Alcohol-based	+ Evaporate faster	- Higher flammability risk - More vapors - Not dielectric
Hydrocarbon-based	+ Dielectric + Less vapors + Lower flammability risk	- Evaporate slower
Fluorinated fluids	+ Dielectric + Evaporates faster + No flammability risk	- Classified as PFAS (potentially restricted use) - High cost
Water-based (not recommended)		- Not dielectric - Evaporate slowly and leave conductive residues

冷卻液性能比較

Fluid type	Mineral Oil	PAO	Silicone oil	Synthetic Ester	Fluorinated oil
Renewable feedstock	+	+	+	+++	+
Biodegradability	+	++	+	+++++	+
Fire safety	++	+++	++++	+++++	+++++
Insulating properties	++++	++++	++++	++++	+++++
Heat transfer capacity	++++	++++	++++	++++	+++++
Oxidation stability	++	++++	++++	+++++	+++++
Low temp rheology	++	+++	++++	++++	+++++

+ Poor, +++++ Excellent

材料相容性

Material Compatibility

Types of Materials

Seals and O-rings/Rubbers

Metals

Sleeving

Plastics

Cables

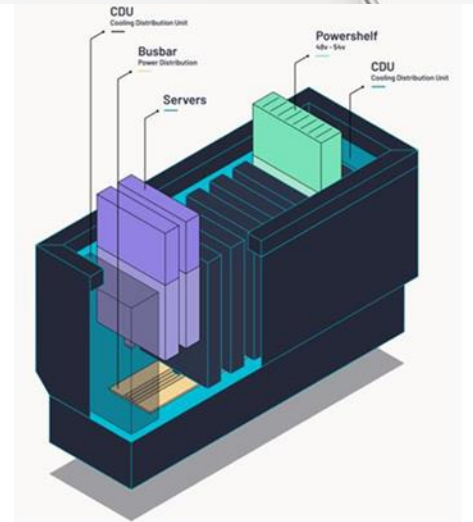
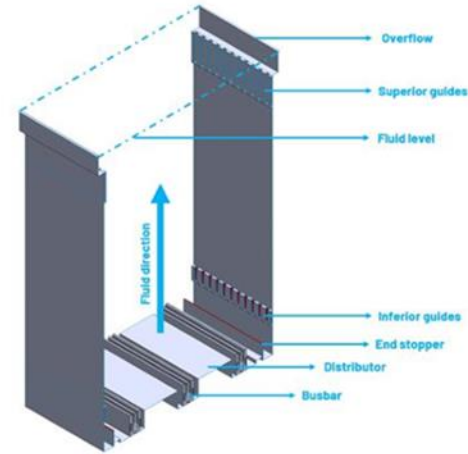
Hose/Piping/Cooling tubes

Other Materials

- Nearly **50 types** of different materials were **evaluated** under these categories.
- **Compatible materials** are identified and commercially available

元件可靠度測試

- 接頭, 連接線及電源及電源母線在浸沒式冷卻中可靠度



浸沒式冷卻技術可靠度需求

- 熱化學品質: 材料不相容性會影響組件及系統中材料的性能
- 熱化學可靠度: 由於箱體及系統材料在操作時產生的汙染造成浸沒組件及系統的性能降低
- 可依循可靠度測試流程進行



FMEA for Thermochemical Failure Modes

Component Material Compatibility

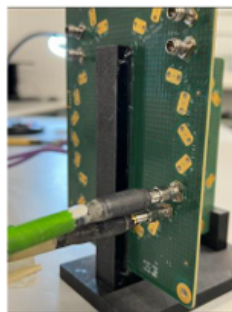
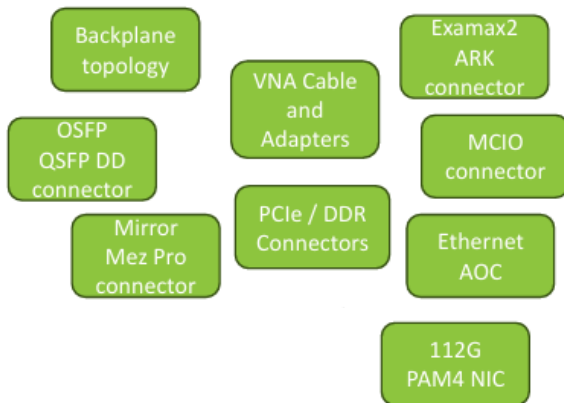
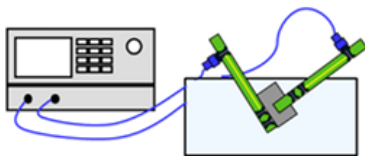
Component Thermochemical Reliability

System Thermochemical Reliability

Reliability Risk Assessment

訊號完整性

- 浸入對 SI 的影響：由於液體介電常數 ϵ_r 和熱環境的變化，眼圖和 BER 也會改變。
- 隨著資料速率的增加，流體與空氣的電磁特性對訊號完整性產生影響。
- 需了解其影響是什麼，以及可以採取哪些措施。



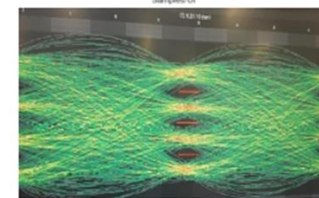
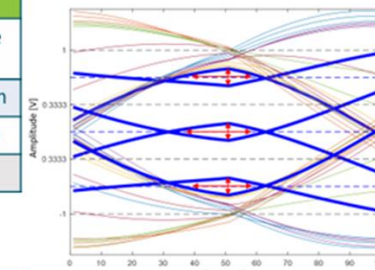
High Speed IO	Immersion Eye compared to Air	Eye Immersion Aging
DDR 5600, 6400MT/s, 8800MT/s	➢ 95 % Small data samples	After 24 months; no change
PCle 5.0 / 4.0	Similar Eyes	After 24 months and 4 month
Ethernet 56G PAM4 per Lane	Pass	After 24 months; no change
Sealed Optical Ethernet	Pass	WIP



Test focus

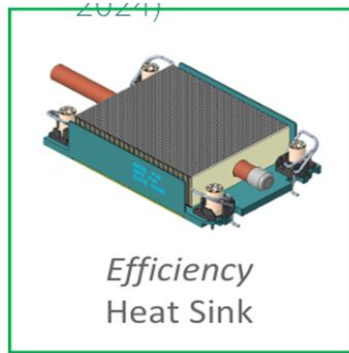
High Speed IO	Compared Air BER with Immersion Eye BER	Worse BER impact to workload Performance
AI/Ethernet 112 Gbps PAM4	* $1e-8 \rightarrow 1e-6$ to $1e-5$	* Need Study BER drop impact to AI workload
AI/Ethernet 224 Gbps PAM4		
PCle 6.0 64 Gbps PAM4	WIP	

PAM4 and BER



浸沒式散熱增強設計

- 結合冷板複合式散熱設計
- 在液體槽中增加流體循環



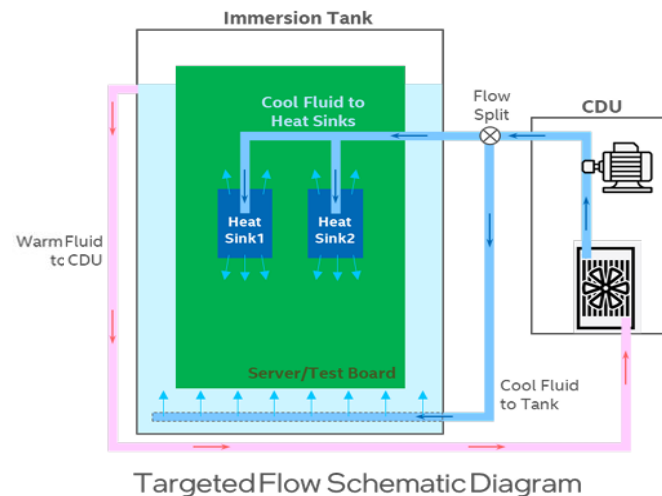
Efficiency
Heat Sink

→ $\Psi_{CL} = 0.055 \text{ } ^\circ\text{C/W}^*$
→ Design Spec Contributed
to OCP 10/24

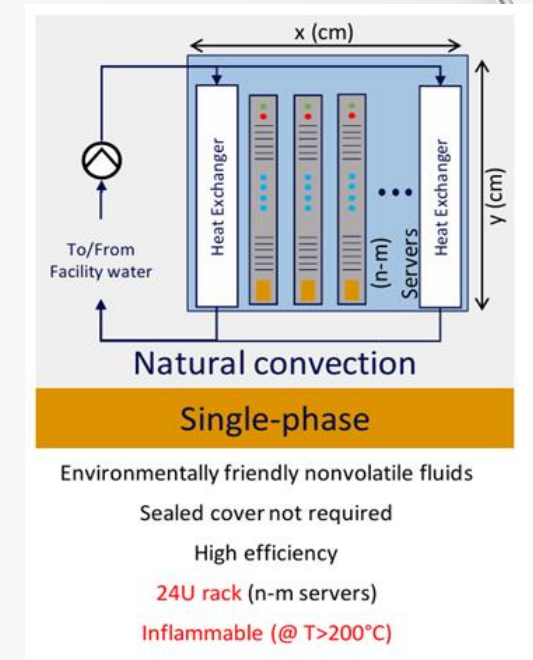


Performance
Heat Sink

→ $\Psi_{CL} = 0.027 \text{ } ^\circ\text{C/W}^{**}$
→ Design available for zero
cost license on Intel platforms

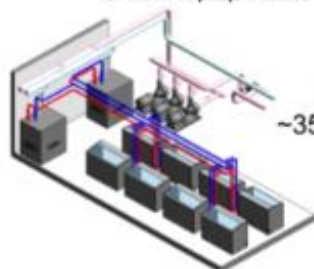


Heat Source	Fluid	Flow	Inlet
*Airport Cove TTV (SP)	PAO 4	2 LPM	40°C
**Orffs Corner TTV (AP)	PAO 4	2 LPM	40°C



熱回收

Data center Heat source
Direct-on-chip Liquid Cooled



~35°C



High-temp Heat pump



~80-105°C "high"

Typical application:
Pasteurization, brewery, F&B
cooking, process hot water,
3rd gen District Heat



~50-80°C "medium"

Typical application:
4th gen District Heat, Fan
coil/AHU, radiator, domestic
water, boiler feedwater,
process wash water, process
pre-heat

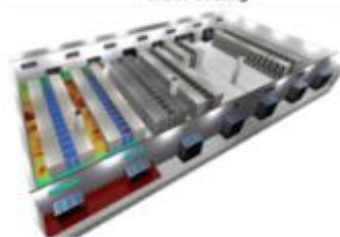


~35-50°C "low"

Typical application:
Greenhouse, underfloor
heating, 5th gen District Heat
(decentralized heat pumps)



CRAC Cooling



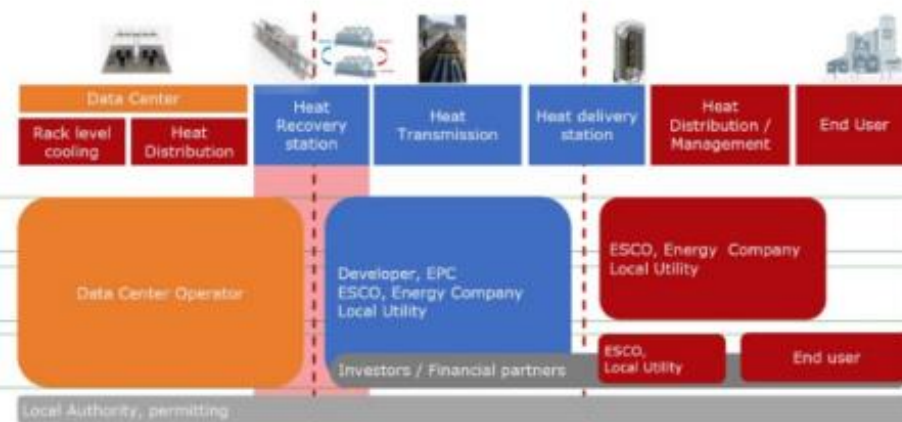
~20°C



Heat pump

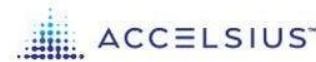


Heat Exchanger



Advanced Liquid Cooling Technology

Direct-to-Chip (D2C) Cooling



Rear Door Heat Exchanger

Passive



Active



(Advanced) Immersion Cooling

One Phase



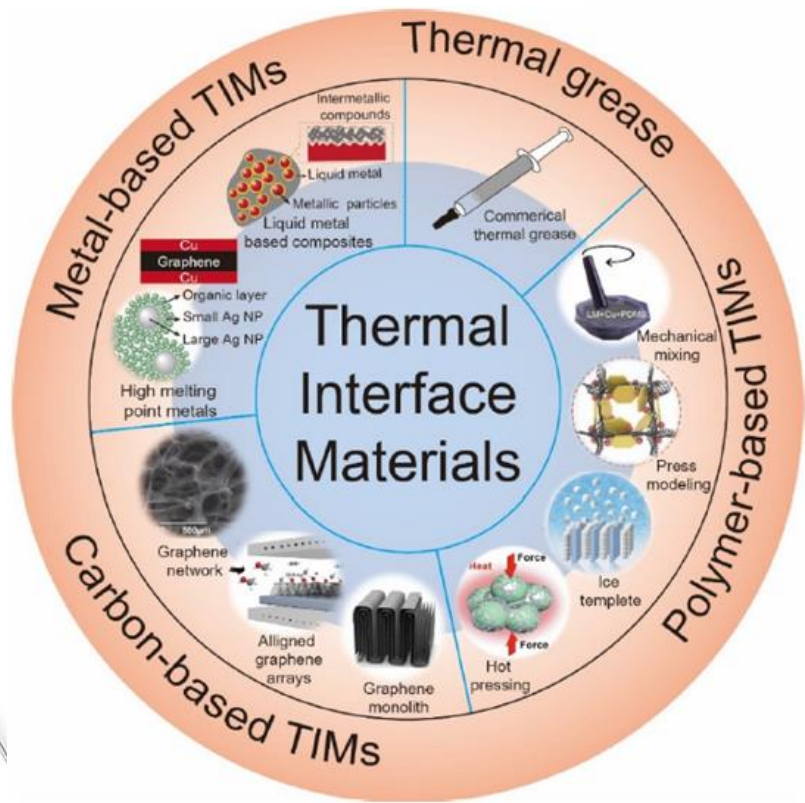
Two Phase



台灣產業機會-冷板加工

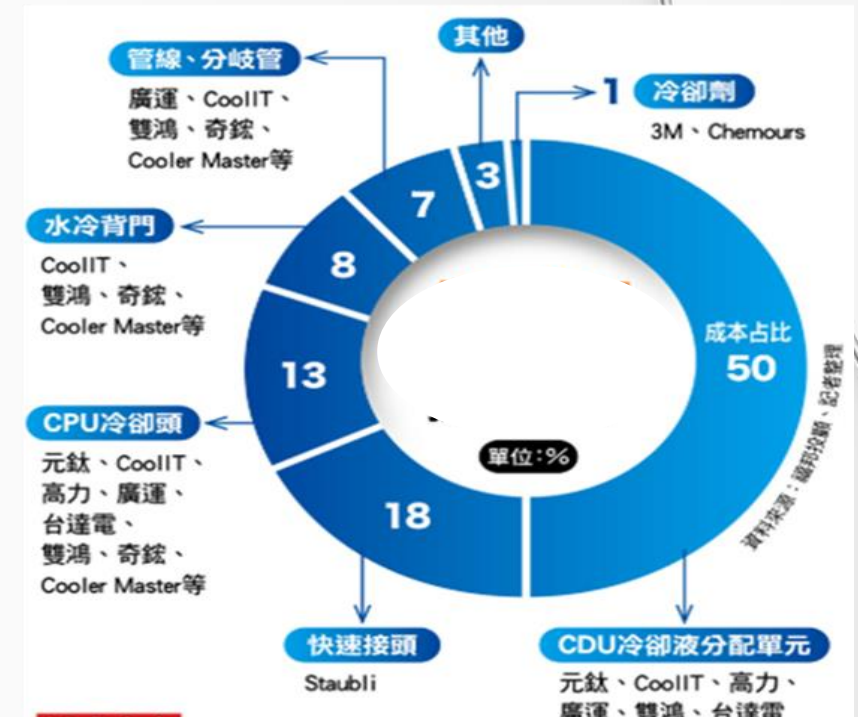
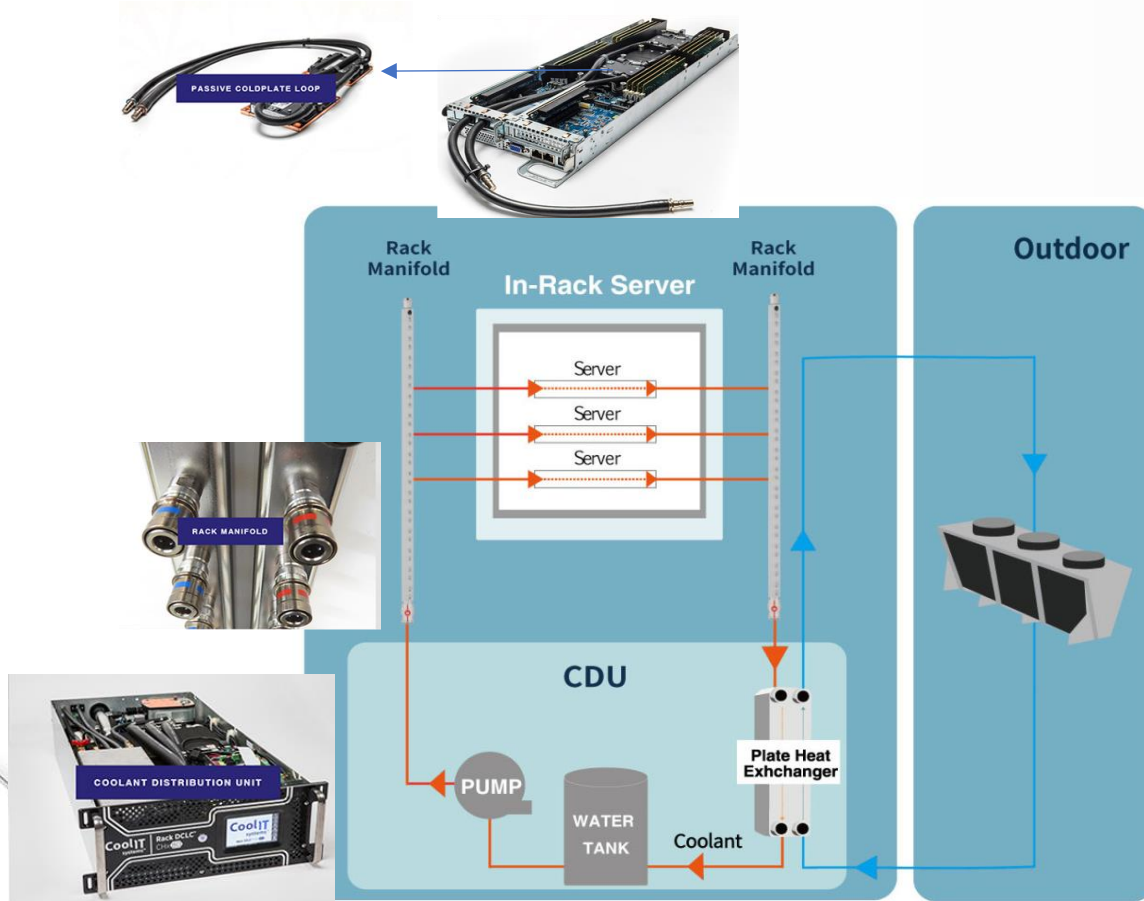
製程	代表可達通道寬度/深度 (典型)	優點
CNC micro-milling	0.2–2 mm	成本可控、量產成熟
Skiving (fin)	fin thickness 100–200 μm , height mm 級	低成本風冷大量生產
EDM (Electrical Discharge Machining)	寬度 100–300 μm , 深度 300–1000 μm	可做到深窄通道、高 aspect-ratio , 耐高硬度材料
L-PBF (SLM / Laser Powder Bed Fusion)	可做通道/管路約 $\geq 300\text{--}500 \mu\text{m}$ (實務)	可做極複雜 3D manifold 與整合通路、快速設計迭代
Binder Jetting + sinter/infiltrate	可達通道 $\sim 200\text{--}800 \mu\text{m}$ (視粒徑與燒結)	可量產複雜形狀、較低成本於 L-PBF、粉床清理較易
Metal 3D + post-machining	300 μm – mm 級	3D 拓撲設計能力強
Electroplating (Cu plating)	通道寬 20–200 μm , 深度 50–300 μm	可直接形成金屬微通道，適合封蓋後成真金屬通道
Laser micromachining	可做到 10–50 μm (小範圍) , 深度類似	高精度、可局部加工複雜形狀
Si etching (DRIE)	寬度 5–200 μm , 深度可至 10 μm – >500 μm	最佳微尺度精度、高 aspect-ratio、光滑側壁、與微加工封蓋/硅-硅鍵合整合性高

台灣產業機會-散熱材料



TIM 種類	物理特性與優勢	導熱性能 (k)	典型應用場合
1. 散熱膏 (Thermal Grease/Paste)	液態/膏狀，不固化。極佳的服貼性 (Wettability)，能形成極薄的黏結層 (低 BLT)，因此界面熱阻最低	中高 (約 1.5 到 10 W/m K)	CPU/GPU核心與散熱器之間。缺點是可能會有滲出 (Pump-out) 問題。
2. 導熱墊片 (Thermal Pad)	固態片狀 (通常為矽膠或彈性體)。具有彈性，能填補較大或不規則的間隙 (例如 0.5 ~5 mm)。提供電氣絕緣。	低到中等 (約 1 到 8 W/m K)	記憶體顆粒 (DRAM)、電源模組 (VRM/MOSFET)、筆電機殼與散熱器等，用於填補高度差異較大的元件。
3. 相變材料 (Phase Change Material, PCM)	常溫下為固態，受熱 (通常 45 ~60 °C) 後融化成液態，服貼性極佳。結合了墊片的易用性與散熱膏的低熱阻優勢。	中高 (約 3 到 15 W/m K)	高效能伺服器、數據中心。首次使用時性能最佳，無需清潔。
4. 液態金屬 (Liquid Metal, LM)	液態金屬合金 (通常是鎳基)。極低的界面熱阻。導熱性能最高。	極高 (約 20 到 80 W/m K)	導熱性能高，缺點是導電，且可能腐蝕鋁材，應用時需謹慎。
5. 導熱膠/灌封膠 (Thermal Adhesive/Epoxy)	液態，會固化。提供永久的機械黏合力。	中等 (約 1 到 5 W/m K)	將小型散熱片黏合到晶片 (例如 LED、晶片組)，或用於灌封電子元件，提供導熱和保護。
6. 導熱石墨片 (Graphite Sheet)	固態片狀 (熱解石墨)。極輕薄，具有極高的平面導熱性 (約 100 到 1500 W/m K)。	平面極高；垂直低	超薄手機、平板電腦。主要用於熱擴散，將熱點的熱量均勻分散到大面積。

台灣產業機會-CDU



結論

- AI伺服器GPU/CPU功能需求，發熱增加，提升帶動新一波液冷散熱技術發展。
- 液冷散熱較氣冷散熱效能顯著提升，組件及設施建置需求提升。
- 直接晶片冷卻採用冷板方式集中晶片端散熱，具有設備建置成本及維修較容易優勢，浸沒式冷卻將晶片及PCB浸沒於介電液中，具有散熱性能優式。
- 單相浸沒式設備建置及冷凝液較雙相單純，雙相浸沒式冷卻有較高的散熱效能。
- 各種液冷散熱方式各有優缺點，散熱效能、可靠度及系統設施建置成本是重要考量。
- 由AI帶動的液冷需求帶來新的散熱市場及機會，台灣廠商可積極投入。

Thank You for Your Listening !

謝謝聆聽！

